

# Program overview

08-Dec-2022 11:18

**Year** 2022/2023  
**Organization** Applied Sciences  
**Education** Bachelor Nanobiology

| Code                                   | Omschrijving                                  | ECTS                                  | p1   p2   p3   p4   p5 |
|----------------------------------------|-----------------------------------------------|---------------------------------------|------------------------|
| <b>Bachelor NB 2022</b>                |                                               |                                       |                        |
| <b>1e jaar BSc NB 2022</b>             |                                               | <b>1st year BSc NB 2022</b>           |                        |
| NB1012                                 | Biochemistry                                  | 3                                     |                        |
| NB1016                                 | Molecular Biology                             | 3                                     |                        |
| NB1022                                 | Genetics                                      | 4                                     |                        |
| NB1031                                 | Introduction to studying Nanobiology          | 3                                     |                        |
| NB1052                                 | Journal Club 1                                | 3                                     |                        |
| NB1072                                 | Physical Biology of the Cell 1                | 3                                     |                        |
| NB1102                                 | Chemistry 1                                   | 3                                     |                        |
| NB1110                                 | Chemistry 2                                   | 3                                     |                        |
| NB1120                                 | Biomolecular Programming                      | 3                                     |                        |
| NB1132                                 | Biophysics                                    | 3                                     |                        |
| NB1140                                 | Physics 1a                                    | 4                                     |                        |
| NB1143                                 | Physics 1B                                    | 3                                     |                        |
| NB1150                                 | Lab Course Track A-1                          | 3                                     |                        |
| NB1151                                 | Lab Course Track A-2                          | 3                                     |                        |
| NB1163                                 | Lab Course Track B-1                          | 3                                     |                        |
| NB1164                                 | Lab Course Track B-2                          | 3                                     |                        |
| NB1201                                 | Analysis 1                                    | 5                                     |                        |
| NB1206                                 | Analysis 2                                    | 5                                     |                        |
| NB1211                                 | Analysis 3                                    | 3                                     |                        |
| NB1230                                 | Linear Algebra                                | 3                                     |                        |
| <b>2e jaar BSc NB 2022</b>             |                                               | <b>2nd year BSc NB 2022</b>           |                        |
| NB2011                                 | Thermodynamics and Transport                  | 3                                     |                        |
| NB2022                                 | Philosophy and Ethics                         | 3                                     |                        |
| NB2032                                 | Evolutionary Developmental Biology            | 6                                     |                        |
| NB2041                                 | Optics and Microscopy                         | 3                                     |                        |
| NB2046                                 | Microscopy / Nanoscopy practice               | 1,5                                   |                        |
| NB2071                                 | Physical Biology of the Cell 2                | 3                                     |                        |
| NB2081                                 | Nanotechnology                                | 2                                     |                        |
| NB2121                                 | Image Analysis                                | 3                                     |                        |
| NB2141                                 | Physics 2                                     | 3                                     |                        |
| NB2151                                 | Journal Club                                  | 1                                     |                        |
| NB2161                                 | Bioinformatics                                | 4,5                                   |                        |
| NB2171                                 | Statistics                                    | 3                                     |                        |
| NB2181                                 | Computational Science                         | 3                                     |                        |
| NB2191                                 | Differential equations                        | 3                                     |                        |
| NB2214                                 | Electronic Instrumentation                    | 6                                     |                        |
| NB2220                                 | Statistical Physics                           | 3                                     |                        |
| NB2230                                 | Biomolecular Structure and Function           | 3                                     |                        |
| TN2545                                 | Systems and Signals                           | 6                                     |                        |
| <b>3e jaar BSc NB 2022</b>             |                                               | <b>3rd year BSc NB 2022</b>           |                        |
| NB3000                                 | Bachelor Thesis Project                       | 20                                    |                        |
| <b>3e jaar BSc NB keuzevakken 2022</b> |                                               | <b>3rd year BSc NB electives 2022</b> |                        |
| NB3011                                 | Nanomedicine                                  | 2,5                                   |                        |
| NB3014                                 | Computational Neuroscience                    | 2,5                                   |                        |
| NB3015                                 | A primer in Neuroscience                      | 2,5                                   |                        |
| NB3016                                 | High Speed Scientific Computing               | 2,5                                   |                        |
| NB3017                                 | Quantum mechanics for Nanobiology 1           | 2,5                                   |                        |
| NB3018                                 | Quantum mechanics for Nanobiology 2           | 2,5                                   |                        |
| NB3019                                 | Molecular Motors                              | 2,5                                   |                        |
| NB3020                                 | Genomics Technology in Breast Cancer Research | 2,5                                   |                        |
| NB3021                                 | Optics and its application in Nanobiology     | 2,5                                   |                        |
| NB3022                                 | Epigenetics                                   | 2,5                                   |                        |
| NB3023                                 | Human Complex Genetics                        | 2,5                                   |                        |
| NB3024                                 | Advanced Math Topics                          | 2,5                                   |                        |
| NB3030                                 | Nanobiology Independent Research              | 0                                     |                        |
| TBM301A                                | Writing a Bachelors Thesis in English         | 2,5                                   |                        |

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| <b>Year</b>         | <b>2022/2023</b>            |
| <b>Organization</b> | <b>Applied Sciences</b>     |
| <b>Education</b>    | <b>Bachelor Nanobiology</b> |

| <b>Bachelor NB 2022</b>                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Director of Education</b>                 | Dr. J. Pothof                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Program Coordinator</b>                   | J.C. Colgrove                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Program Coordinator</b>                   | D. Mevius                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Contact for students</b>                  | Info-BSc-NB <Info-BSc-NB@tudelft.nl>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Program Title</b>                         | BSc Nanobiology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>In association with the University of</b> | Erasmus Medical Center in Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>EC Program</b>                            | <p>The programme has a study load of 180 credits. This includes the first academic year, with a study load of 60 credits, which is concluded with a binding recommendation on the continuation of studies. The second and third academic years have a combined study load of 120 credits. This phase includes a minor with a study load of 30 credits. The programme includes the Bachelors degree audit.</p> <p>The programme consists of</p> <p>Year 1: 60 EC, courses in math, physics, biology, chemistry, integrated courses, practical courses and academic skills</p> <p>Year 2: 60 EC, courses in math, physics, biology, integrated courses, practicals and academic skills</p> <p>Year 3: 30 EC minor, 20 EC Bachelor end project, and 10 EC of nanobiology electives.</p> <p>As a component of the programme, the minor includes the following variants</p> <ol style="list-style-type: none"> <li>Thematic minor, as approved by the university,</li> <li>Self-composed minor, as approved by the Board of Examiners.</li> </ol> <p>The Bachelors degree programme is concluded with a bachelor end project. This project demonstrates that the student possesses and is able to apply the knowledge, insight and skills acquired in the degree programme.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Program Goals</b>                         | <p>The programme is intended to educate students to earn a Bachelor of Science degree in Nanobiology, providing them with such a level of knowledge, insight and skills in the area of Nanobiology, that graduates can fulfil positions on the labour market at the Bachelors level.</p> <p>And be eligible for a follow-up programme at the Masters level, in any case, the Masters degree programme(s) in Nanobiology at TU Delft/Erasmus Medical centre, Infection and Immunity, Molecular Medicine and Neuroscience at Erasmus Medical centre.</p> <p>Graduates must also meet the specific final attainment levels for each degree programme. The Bachelor of Science graduate of the programme Nanobiology that has the aim to bridge the fields of mathematics, physics and biology. The student:</p> <ol style="list-style-type: none"> <li>will have relevant, current and fundamental knowledge of mathematics, physics and biology, as well as the methods and techniques of scientific research.</li> <li>will be able to identify related concepts in biology and physics and to apply knowledge from one field of science to another.</li> <li>will be able to use the acquired knowledge to follow current scientific research in the fields of biology and biophysics intensively, in addition to understanding and interpreting this literature.</li> <li>will have demonstrable experimental research skills in the fields of molecular biology and biophysics.</li> <li>will have the required communication skills.</li> <li>will have an understanding of the ethical issues surrounding scientific research.</li> <li>will be aware of the need for lifelong learning and of the utility of creativity to the achievement of scientific progress.</li> </ol> |
| <b>Gives access to</b>                       | <p>Students who have successfully completed their Bachelor's degree programme and have therefore been granted the title Bachelor of Science in Nanobiology (joint degree) are in any case admissible to the following Master of Science degree programmes: Nanobiology joint programme with TU Delft and Erasmus University Rotterdam, and to Infection and Immunity; Molecular Medicine; and Neuroscience at Erasmus University Rotterdam</p> <p>Access to other Master of Science degree programmes is subject to the conditions stipulated by the institute that provides that degree programme.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Expected prior Knowledge</b>              | <p>Graduates of the Bachelor programme enter many different masters programmes.</p> <p>Access requirement(s), applicants must meet one of these criteria:</p> <p>A VWO (6 years of pre-university education) certificate with the subject cluster 'Nature and Technology' or with the subject cluster 'Nature and Health' but only if Mathematics B, Biology, Physics and Chemistry were taken as VWO examination subjects.</p> <p>A propedeutic certificate of a university degree programme (WO)</p> <p>A degree certificate from a University of Applied Science (HBO):</p> <p>Applicants with a VWO certificate: see the VWO subject cluster requirements.</p> <p>Applicants with a HAVO (5 year of pre university education) or</p> <p>MBO (intermediate vocational education) certificate must have filled all gaps in knowledge compared to the VWO level in the following subjects: Mathematics B, Biology, Physics and Chemistry. On individual basis, proof of English proficiency can be required.</p> <p>Other qualifications recognised by the university's Executive Board which may or may not have been obtained in the Netherlands and are seen as being at least equivalent to a VWO examination certificate holding the subjects Mathematics, Biology, Physics and Chemistry on a proper level.</p> <p>On the basis of a colloquium doctum (i.e. a special entrance examination). In order to sit a colloquium doctum candidates must be at least 21 years of age.</p> <p>All other qualifications should have a level of education equivalent to the VWO level in the subjects Mathematics B, Biology, Physics and Chemistry.</p>                                                                                                                              |

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| <b>Organization</b> | <b>Applied Sciences</b>     |
| <b>Education</b>    | <b>Bachelor Nanobiology</b> |

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|-------------------------------------|---------------|
| <b>1e jaar BSc NB 2022</b>          |               |
| <b>Responsible Program Employee</b> | J.C. Colgrove |
| <b>Director of Education</b>        | Dr. J. Pothof |
| <b>Program Coordinator</b>          | J.C. Colgrove |
| <b>Program Coordinator</b>          | D. Mevius     |

| NB1012                                 | Biochemistry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3 |
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| <b>Responsible Instructor</b>          | Dr. B.C. Tilly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Responsible Instructor</b>          | Dr. G.E. Bokinsky                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr. B.C. Tilly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Dr. G.E. Bokinsky                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/8/0/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 2<br>2B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 2B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>Biochemistry represents the interface between chemistry and biology and studies life at a molecular level. Over the years our knowledge about the molecular composition of cells and the complex network of chemical reactions that take place within them has been vastly increased and is still expanding rapidly. Nowadays, biochemistry plays an important role in the major developments in agriculture, (bio-)technology and health care. In this course you will learn the biochemical principles that underlie cellular physiology. In addition to the molecular composition of cells and the major metabolic pathways involved, special emphasis will be on the structure and function of enzymes and bioenergetics. The knowledge obtained during this course will be applied in other biologically orientated courses, including Lab course Nanobiology, Biophysics, Physical Biology of the Cell and Evolutionary &amp; Developmental Biology. For successful completion of this course, you are expected to be familiar with the concepts and skills of the courses Analysis I and Chemistry I &amp; II.</p> |   |
| <b>Study Goals</b>                     | <p>At the completion of this course you will be able to:</p> <ul style="list-style-type: none"> <li>describe and recognize the major molecular constituents of the cell, their function and how they are synthesized from precursors.</li> <li>describe the modular organization of proteins, how they fold into 3D structures and can form (multifunctional) complexes.</li> <li>predict how activators and inhibitors affect the rate of the reaction using the principles of enzymatic catalysis.</li> <li>Calculate the rate of enzymatic reactions using Michaelis-Menten theory.</li> <li>Predict the effects of specific mutations on enzyme catalysis.</li> <li>know the major metabolic pathways in bacteria and cells and explain how these pathways are regulated to maintain homeostasis.</li> <li>describe photosynthesis in bacteria and plants and how energy is stored in reduction equivalents and ATP.</li> <li>describe how mitochondria convert the energy stored in reduction equivalents into ATP and how the ADP/ATP ratio regulates cellular metabolism.</li> </ul>                                  |   |
| <b>Education Method</b>                | <p>This course contains lectures, tutorials and practicals. During the practicals the properties and kinetics of enzymes will be studied using computer simulations and a laboratory experiment (Practical Alkaline phosphatase, week 3). Attendance of the practical Alkaline phosphatase is mandatory.</p> <p>Due to coronavirus regulations, some changes may be made in the schedule and educational methods used.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Literature and Study Materials</b>  | <p>Molecular Biology of the Cell, Sixth edition, Alberts et al., Garland Science, 2015<br/> Achieve Macmillan learning (subscription required, provided by the programme for Nanobiology students).<br/> Additional learning materials (textbook extracts) will be posted to Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                      | <p>The final grade is based on a written exam and points earned for sufficiently completed home assignments (up to 0.5 points, maximum grade is 10). The bonus points will be added to your grade only if you pass the written exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Remarks</b>                         | <p>Ben Tilly can be reached at b.tilly@erasmusmc.nl</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Studyload/Week</b>                  | <p>The course (5 weeks) consists of lectures, online and in class tutorials (32h), practical work (4h), home assignments (8h) and 40h of study at home and exam preparation.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Location</b>                        | Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| NB1016                                 | Molecular Biology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof. W. Vermeulen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Responsible Instructor</b>          | A. Ray Chaudhuri                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | <p>Molecular Biology focuses on</p> <ol style="list-style-type: none"> <li>1) storage and transfer of information in biological systems, from DNA to mRNA to protein, and</li> <li>2) the various mechanisms by which the integrity of DNA is maintained.</li> </ol> <p>Throughout the course there will be specific focus on structural requirements, as well spatial and temporal aspects of the specific pathways.</p> <p>NB1016 provides basic knowledge that is required in a number of subsequent related courses, including Physical Biology of the Cell, Biophysics, and Evolutionary &amp; Developmental Biology.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Study Goals</b>                     | <ol style="list-style-type: none"> <li>1. Describe the structure of DNA</li> <li>2. Describe the anatomy of a DNA replication fork and function of replication proteins</li> <li>3. Explain how accuracy in DNA replication is achieved</li> <li>4. Explain the end replication problem</li> <li>5. Describe how transcription is initiated using general transcription factors</li> <li>6. Describe how different classes of transcription factors interact with DNA</li> <li>7. Explain how transcription can be regulated using gene regulatory proteins</li> <li>8. Explain how proteins are synthesized through translation of mRNA</li> <li>9. Describe how proteins are degraded by the proteasome</li> <li>10. Describe at the basic level DNA repair pathways and the lesions they act on</li> <li>11. Understand the link between DNA recombination and replication</li> <li>12. Describe the different phases of the cell cycle</li> <li>13. Understand how the transitions between the different cell cycle phases occur</li> <li>14. Explain how cell cycle checkpoints are established</li> </ol> |   |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>- Lectures</li> <li>- Individual/group assignments</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | 1 exam counts for 100%. Due to yet unforeseen developments due to COVID-19 the way the course is assessed might change.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Remarks</b>                         | <p>Wim Vermuelen can be reached at w.vermeulen@erasmusmc.nl</p> <p>Arnab Ray Chaudhuri can be reached at a.raychaudhuri@erasmusmc.nl</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Studyload/Week</b>                  | <p>7h/week * 8 = 56 h, including lectures, problem sessions and self study</p> <p>28 h exam preparation</p> <p>Total of 84 hours</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Location</b>                        | Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| NB1022                                 | Genetics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4 |
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| <b>Responsible Instructor</b>          | Dr. W.J. Lans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Responsible Instructor</b>          | Dr.ir. J.A.F. Marteiijn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 6/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 1A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | In this course the student will learn the principles of genetics, ranging from classical Mendelian genetics to the most recent molecular genetics techniques. Both the theoretical and practical aspects of genetics will be taught, including DNA/RNA structure, gene function, genome organization, inheritance, DNA replication and recombination and mutation. A major part of this course will cover various classical and modern molecular genetic techniques from the genomics era. The course includes individual and group exercises focusing on solving genetic problems. The knowledge gained in this course will show the student how to design and interpret genetic experiments, to answer relevant genetic questions. Students will need this knowledge for other courses including Molecular Biology, Laboratory Practical, Physical Biology of the cell, Evolutionary and Developmental Biology; and Evolution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                     | <p>At the end of this course, the student will be able to:</p> <ul style="list-style-type: none"> <li>describe the function and organization of DNA in prokaryotes and eukaryotes (building blocks, genes, regulatory elements, gene layout)</li> <li>formulate the basic principles of transcription and translation (genetic code, RNA and protein building blocks, splicing, non-coding RNA, RNA, layout, regulation of transcription and translation)</li> <li>describe the function and organization of chromatin (histones, nucleosomes, chromosomes, centromeres and telomeres, eu- and heterochromatin, basic epigenetics)</li> <li>understand and explain the concepts and terminology of classical genetics and inheritance (mendelian inheritance, dominance and recessiveness, homo- and heterozygote, X-linked, pedigrees, linkage, allele, population genetics)</li> <li>describe how genomes are copied and transmitted (replication, mitosis, meiosis, recombination, linkage analysis)</li> <li>understand and describe different types of mutations, their effects and chromosome instability (missense and nonsense mutations, deletion/insertion, LOH, translocation, CIN, transposons)</li> <li>list causes of different genetic diseases and their therapies (gene therapy, genetic screening, hereditary syndromes, cancer, trinucleotide repeat disorder)</li> <li>describe differences between the genetics of eukaryotes, prokaryotes and viruses (RNA/DNA viruses, bacteria)</li> <li>understand both classical and modern molecular genetic techniques (genetic analysis, complementation analysis, mapping analysis, cytogenetics, forward/reverse genetic experiments, sequencing, PCR, expression analysis, microarrays, quantitative PCR, hiseq, CGH/tiling arrays, genomics)</li> <li>interpret and critically evaluate results from genetic experiments</li> <li>design genetic experiments to study the function of genes and their effect on phenotype, both individually as well as in groups (knockout, siRNA technology, recombinant DNA, cloning, expression and siRNA libraries)</li> </ul> |   |
| <b>Education Method</b>                | <p>Lectures</p> <p>Group Assignments (interpreting and designing genetic experiments)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Literature and Study Materials</b>  | <p>Genetics, a conceptual approach, 7th edition<br/>by Benjamin A. Pierce (Palgrave Macmillan/W.H. Freeman)</p> <p>And readers for assignments that will be distributed via Brightspace</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>1 test (30%)</p> <p>Final exam (60%)</p> <p>Presence and active participation during assignments (10%)</p> <p>In case of a re-exam, the grade for presence and active participation during assignments (10%) will remain and the resit of the exam will replace the grades of the test and the final exam and will count up to 90% of the final grade.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Remarks</b>                         | <p>Responsible Instructors:<br/>Prof. Dr. Ir. Jurgen Marteiijn and Dr. Hannes Lans</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Studyload/Week</b>                  | <p>35 hours of lectures and group assignments for the total course of 4 ECTS. This indicates that approximately 80 hours of self-study is needed during the course.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Location</b>                        | Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

| NB1031                                 | Introduction to studying Nanobiology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. J. Pothof                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 9/0/0/0/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 1<br>1A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Start Education</b>                 | 1<br>1A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | 1<br>1A<br>1B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | In this course the students learn basic study skills and become familiar with the different forms of instruction and work (lectures, practical work, group work, problem sessions). Orientation sessions are aimed at providing group identity and cohesion for students in this new program. Presentations by research faculty introduce students to current topics in Nanobiology. The students become familiar with team based learning, are divided into groups and practice this form of learning to complete a project on a topic in Nanobiology. The students will learn information processing skills; gathering information on a specific topic from the internet, evaluating the value of the information, summarizing information and using the gathered information in a group project. |   |
| <b>Study Goals</b>                     | At the completion of this course the student will be able to:<br>1. reflect on own academic and study skills and pitfalls<br>2. Work effectively in groups with respect for group members<br>3. Critically evaluate and provide positive feedback for working partners<br>4. Find basic scientific information on the internet<br>5. Understand the role of the different topics in the Nanobiology curriculum<br>6. Be familiar with, and adhere to, rules and regulations regarding academic integrity                                                                                                                                                                                                                                                                                            |   |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>- Guided instruction and (short) lectures</li> <li>- Short writing assignments</li> <li>- Team based learning exercises</li> <li>- Group Assignments</li> </ul> <p>For 5 Week block scheduling this course will consist of 19 sessions of mixed short lecture / instruction and (group) work assignments. Expect about 9 active working hours a week and equivalent out of class preparation time. This is subject to change due to relevant Covid19 regulations.</p>                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | Reading material to be handed out or posted on Brightspace including: articles on Personal profile and learning styles, Effective communication and giving/receiving feedback, Working in groups (team roles, problem solving techniques), Team Based Learning and Presentation skills.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Assessment</b>                      | <p>Grade will be pass / not pass</p> <p>There are 5 elements to evaluation and ALL have to be COMPLETED SATISFACTORY to pass.</p> <ol style="list-style-type: none"> <li>1. Complete all short writing assignments(individual)</li> <li>2. Chemistry prior knowledge test and reflection assignment (individual)</li> <li>3. Peer evaluation of group work (individual)</li> <li>4. Final project (report, poster, pitch video)(group)</li> <li>5. attendance and active participation poster session (individual)</li> </ol>                                                                                                                                                                                                                                                                       |   |
| <b>Remarks</b>                         | Joris Pothof can be reached at j.pothof@erasmusmc.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Studyload/Week</b>                  | <p>For 5 Week block scheduling this course will consist of 19 sessions of mixed short lecture / instruction and (group) work assignments. Expect about 9 active working hours a week and equivalent out of class preparation time. This is subject to change due to relevant Covid19 regulations.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                        |   |
| <b>Location</b>                        | Erasmus MC, Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| NB1052                                 | Journal Club 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3 |
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| <b>Responsible Instructor</b>          | Dr. I.V. Smal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Responsible Instructor</b>          | G. Roshchupkin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | In Journal Club several scientific research skills will be taught and practiced. Using examples from scientific literature and popular press the students become acquainted with current research in biomedicine, nanoscience, biophysics and related fields. Introductory sessions will focus on how to find scientific information in print and electronic form, the various types of information sources, how to evaluate the worth of different information sources and specifically how to read a scientific research article. Working sessions will be team based learning where each student group will be given an article to read and a set of questions to answer based on the article. Finally student groups will give a presentation on the article they have read and relevant background articles, based on assigned questions. This course utilizes and builds on team learning skills introduced in Intro to Nanobiology. The skills of processing scientific and research information are required for subsequent courses. |   |
| <b>Study Goals</b>                     | <p>At the completion of this course the student will be able to:</p> <ol style="list-style-type: none"> <li>1. identify different information sources commonly used in scientific research and describe their characteristics and use.</li> <li>2. identify the type of information found in the different parts of a research article (abstract, introduction, materials and methods, results, discussion)</li> <li>3. use internet resources to search for and retrieve research articles, and related background information.</li> <li>4. work effectively as a team to answer a set of specific questions and create a presentation</li> <li>5. constructively evaluate peer group members with respect to contribution to group assignments</li> <li>6. identify alternative explanations for aspects of published research as a step toward critical evaluation</li> <li>7. discuss the difference between scientific information presented in research articles and in the popular press.</li> </ol>                                  |   |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>- Short introduction and instruction lectures</li> <li>- Meetings of groups to work on assignments, team-based learning</li> <li>- Group presentation</li> </ul> <p>This course will consist of sessions including short instructions and group work activities, with variable amounts of additional preparation and reading required.<br/>NOTE: we will monitor current policy developments and regulations to decide whether in-person gatherings for groupwork and on-campus instruction will be possible.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | <p>This is a graded course.<br/>Final grade is determined as follows:<br/>90% final group presentation<br/>10% based on peer feedback on contributions to group work<br/>Further, students have to pass an online information literacy course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Studyload/Week</b>                  | Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Location</b>                        | Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |



| NB1072                                 | Physical Biology of the Cell 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3 |
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| <b>Responsible Instructor</b>          | Dr. M.W.J. Fornerod                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>The cell is the basic unit of biology. One of the important goals in the nanobiology curriculum is to prepare students for constructing artificial cells. How to do exactly do this we don't know, because artificial cells have never been created. In this course we will identify the cellular functions which make the cell different from other objects in particular self replication, evolution, and differentiation by genetic networks. These topics will be studied in an abstract, almost philosophical manner. Students will gain hands on understanding in these processes using computer model systems including self replicating cellular automata and boolean networks. The course includes presentations of the biological reality and aims to make the connection between this reality and the more abstract cellular principles which they represent.</p> <p>The course will assume knowledge of cellular biomolecules gained in the courses of Chemistry, Genetics and Molecular Biology. This course is helpful for understanding the underlying mechanisms of Evolutionary &amp; Developmental Biology, and connects to Biomolecular Structures, and signalling pathway biology presented in Physical Biology of the Cell 2.</p> <p>Session 1 The Minimal Cell<br/>In this session we're going to discuss the concept of the cell, the fundamental building block of life, and what a minimal cell could be. In the practical we're exploring Conway's game of life.</p> <p>Session 2 A Cell from Scratch<br/>We'll make a cell from scratch making use of its capability to self-replicate. In the practical we'll analyze John Von Neumann's universal constructor and work with self replicating loops using the Golly program.</p> <p>Session 3 Cellular evolution from Scratch<br/>One of the important functional requirements of the cell which we have discussed in session 1 (and 2) was "to evolve". Having build a cell from scratch, we now want to investigate whether it is possible to create cellular evolution from scratch. The big advantage could be that, what usually takes years if not centuries, if not millions of years, we could be analyzing in real time every step of the way.</p> <p>Session 4+5 Cells integrate signals through networks to obtain dynamic stability<br/>In this session we discuss how signals from the outside of the cell are integrated and communicate with the genome. We'll discuss how cells manage to integrate signals that come in from the environment to decide what a cell is going to do next. In the practical, we'll simulate simple and more complex networks using the R package BoolNet. There will be an introduction to R included in this session. A home assignment connected to this practical is to be uploaded.</p> <p>Sessions 6+7 Genetic networks underlie cellular differentiation<br/>In this part of the PBC course we'll take a look at perhaps the greatest achievement of cells, their capacity to differentiate and self-organize into complex (even thinking) structures. Genetic networks underlie this differentiation process, which we will probe by looking at the human hematopoietic system. We will also discuss stem cells and stem cell reprogramming, for which two 2012 Nobel prizes were awarded.</p> <p>Session 8 Artificial life, from computational to lab experiments. To what extent does our knowledge of the fundamental properties of the cell help us in the engineering of an artificial cell? What is the current state of the art in making an artificial cell from scratch in the lab?</p> |   |
| <b>Study Goals</b>                     | <p>Upon completion of this course the student will be able to:</p> <ol style="list-style-type: none"> <li>1. Identify and construct several of the most important characteristics of the cell in silico: self replication, cellular evolution, homeostasis by signaling networks, and differentiation by genetic networks.</li> <li>2. Connect the biological reality of the cell to these more abstract principles.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | <p>Lectures<br/>Reading/viewing assignments<br/>Math modeling assignments<br/>Computer modeling assignments<br/>Experimental design assignments</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | <p>Specific information on each session including reading materials will appear on BB announcements.</p> <p>Reading materials and how to prepare for the exam.</p> <p>Books used in the course are: Alberts et al., Molecular Biology of the Cell, 6th edition.</p> <p>In this course, the lectures and other in class activities are the leading content. No more detail is required to be learned for the exam than presented in the lectures, the required reading and practicals. Use the textbooks and other (background) reading materials as a reference in order to understand the lectures before or after the lectures or before the exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Assessment</b>                      | <p>The practical is an integral part of the session and should not be skipped or left early. The practical assignments are not graded but have to be uploaded. Making them is essential in order to successfully complete the exam, because assignment performance is tightly correlated to grade performance.</p> <p>As a bonus, across the eight sessions there will be the possibility to earn 1 bonus point that together translate into one full grade point. The possibility to obtain bonus points, and how, will be announced ahead of the session in which they can be awarded. These bonus points would in principle allow you to obtain a grade of 10.75. However, all grades higher than 10 are rounded off to 10. Bonus points are carried over to the resit.</p> <p>A practice exam will be provided at least 1 week before the exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Remarks</b>                         | <p>Responsible instructor:<br/>Dr. Maarten Fornerod (m.fornerod@erasmusmc.nl)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Studyload/Week</b>                  | <p>Study load is 3.5h of in class education per week (8 weeks) plus assignment and self study work, also approximately 3.5h a week. Time for preparation for the exam depends very much on in class commitment.<br/>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

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|-----------------|-----------|
|                 | you.      |
| <b>Location</b> | Rotterdam |

| <b>NB1102</b>                          | <b>Chemistry 1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>3</b> |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Responsible Instructor</b>          | Dr. B.C. Tilly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
| <b>Instructor</b>                      | Dr. J.A.A. Demmers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>Instructor</b>                      | Dr. J.A. van der Knaap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/6/0/0/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>Education Period</b>                | 1<br>1B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>Start Education</b>                 | 1B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>Exam Period</b>                     | 1<br>2<br>1B<br>2A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>Course Contents</b>                 | <p>Chemistry studies the composition, properties and transformations of matter. Chemical principles apply to living organisms and knowledge of chemistry is essential for the understanding of biological processes. Understanding chemistry is especially important for research at the nanoscale, where complex interactions between different molecules are governed by chemical forces and barriers.</p> <p>In this course the student will learn the fundamental properties of atoms and molecules, how they react with each other and obey the laws of thermodynamics. In addition the student will obtain basic knowledge of analytical and spectroscopic techniques used in life sciences. The student is expected to be familiar with the concepts and skills of Math as taught during the course Analysis 1. The knowledge obtained during this course will be applied in other biologically orientated courses including Biochemistry, Biophysics, Evolutionary &amp; Developmental Biology, Lab Practicum and Physical Biology of the Cell.</p>                                                                                                                                      |          |
| <b>Study Goals</b>                     | <p>At the completion of this course the student should be able to:</p> <ul style="list-style-type: none"> <li>- Understand the structure and properties of atoms in terms of quantization, wavefunctions, atomic orbitals and periodicity.</li> <li>- Describe the characteristics of the Lewis, valence bond, and molecular orbital theories of chemical bonds</li> <li>- Describe the characteristics of ionic, covalent and metallic bonding and predict the type of bonding from the electronegativities of the atoms</li> <li>- Recognize non-covalent interactions and predict physical properties of atoms and molecules.</li> <li>- Predict the shape of molecules using valence shell electron pair repulsion theory (VSEPR)</li> <li>- Construct molecular orbital energy level diagrams for diatomics</li> <li>- Recognize functional groups in organic molecules and name these compounds using IUPAC nomenclature and common names.</li> <li>- Know the Arrhenius, Brønsted-Lowry and Lewis definitions of acids and bases and calculate the pH of solutions of acids, bases and buffers</li> <li>- Describe and predict the behaviour of gases using the ideal gas law.</li> </ul> |          |
| <b>Education Method</b>                | <p>This course contains lectures, tutorials and home assignments.</p> <p>Due to coronavirus regulations, some changes may be made in the schedule and educational methods used.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Literature and Study Materials</b>  | <p>Chemistry3. Fourth edition. Burrows, Holman, Parsons, Pilling and Price, Oxford University Press, 2021.</p> <p>Achieve Macmillan learning (subscription required).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>Assessment</b>                      | <p>Final grade for the course is based on the exam plus 0.1 point bonus for each sufficiently completed home assignment (max. 0.7 points). The bonus will be received only once and will be added to your grade the first time the exam is taken (maximum grade is 10).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Remarks</b>                         | <p>Ben Tilly can be reached at <a href="mailto:b.tilly@erasmusmc.nl">b.tilly@erasmusmc.nl</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Studyload/Week</b>                  | <p>The course (5 weeks) consists of lectures, online and in class tutorials (35h), home assignments (11h) and 38h of study at home and exam preparation.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Location</b>                        | Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |

| NB1110                                 | Chemistry 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3 |
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| <b>Responsible Instructor</b>          | Dr. B.C. Tilly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. J.A.A. Demmers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Instructor</b>                      | Dr. J.A. van der Knaap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/8/0/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 2<br>2A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                     | 2A<br>2B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | <p>Chemistry studies the composition, properties and transformations of matter. Chemical principles apply to living organisms and knowledge of chemistry is essential for the understanding of biological processes. Understanding chemistry is especially important for research at the nanoscale, where complex interactions between different molecules are governed by chemical forces and barriers.</p> <p>In this course the student will learn the fundamental properties of atoms and molecules, how they react with each other and obey the laws of thermodynamics. In addition the student will obtain basic knowledge of analytical and spectroscopic techniques used in life sciences. The student is expected to be familiar with the concepts and skills of Math as taught during the course Analysis 1. The knowledge obtained during this course will be applied in other biologically orientated courses including Biochemistry, Biophysics, Evolutionary &amp; Developmental Biology, Laboratory Practicum, and Physical Biology of the Cell.</p>                                                                                                                                                   |   |
| <b>Study Goals</b>                     | <p>At the completion of this course the student will should be able to:</p> <ul style="list-style-type: none"> <li>- Describe and explain the First Law of Thermodynamics and calculate the enthalpy and internal energy changes for a chemical reaction</li> <li>- Describe and explain the Second Law of Thermodynamics and calculate the entropy and Gibbs energy changes of a reaction</li> <li>- Define and use the thermodynamic equilibrium constant and know how it depends on reaction conditions</li> <li>- Know what is meant by the rate of the reaction and how it depends on reactant concentrations, temperature and the presence of catalysts, including enzymes</li> <li>- Recognize structural isomers and stereoisomers</li> <li>- Understand the basic principles of molecular spectroscopy in terms of quantization of molecular energy and transitions between molecular energy levels when matter interacts with radiation</li> <li>- Describe electrochemical, chromatographic and spectroscopic methods of analysis and suggest suitable methods for a given analyte.</li> <li>- Determine the structure of an unknown compound by using mass spectra, IR spectra and NMR spectra</li> </ul> |   |
| <b>Education Method</b>                | <p>This course contains lectures, tutorials and home assignments.</p> <p>Due to coronavirus regulations, some changes may be made in the schedule and educational methods used.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Literature and Study Materials</b>  | <p>Chemistry3. Fourth edition. Burrows, Holman, Parsons, Pilling and Price, Oxford University Press, 2021.</p> <p>Achieve Macmillan learning (subscription required).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Assessment</b>                      | <p>Final grade for the course is based on the exam plus 0.1 point for each sufficiently completed home assignment (max. 0.7 points). The bonus will be received only once and will be added to your grade the first time the exam is taken (maximum grade is 10).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                         | <p>Ben Tilly can be reached at <a href="mailto:b.tilly@erasmusmc.nl">b.tilly@erasmusmc.nl</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Studyload/Week</b>                  | <p>The course (5 weeks) consists of lectures, online and in class tutorials (35h), home assignments (11h) and 38h of study at home and exam preparation.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Location</b>                        | Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

| NB1120                                 | Biomolecular Programming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. A.B. Houtsmuller                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Responsible Instructor</b>          | Dr. W.A. van Cappellen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4<br>4A<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>In the course Biomolecular Programming (BMP), students learn how to design algorithms for performing specific tasks and to implement algorithms in simple and advanced computer programs in the programming language JAVA. They learn how to use the acquired programming skills to design and implement Computer simulation programs to simulate the behaviour of biomolecules in the cell and the cell nucleus.</p> <p>BMP is complementary to various topics in molecular biology and related courses since the insight in cellular dynamics obtained by simulating and visualising dynamic behaviour of biomolecules provides a strong basis to understand how biophysical and biochemical principles apply to the function of the living cell, or living systems. BMP links to the Analysis and Differential Equations courses, where skills are taught required to establish analytical models to describe the modeled systems. BMP also links to Physical Biology of the Cell since the simulation software package BSL, is largely based on algorithms implemented by the students during the course.</p>             |   |
| <b>Study Goals</b>                     | <p>At the completion of this course the student will be able to:</p> <ol style="list-style-type: none"> <li>1. Demonstrate understanding of the concept ALGORITHM (in the strict meaning)</li> <li>2. Design flow diagrams to schematically describe the subsequent steps and loops within an ALGORITHM</li> <li>3. Demonstrate the capacity to translate (=implement) an ALGORITHM into a (part of a) COMPUTER PROGRAM, including the specific skills to use in JAVA: <ul style="list-style-type: none"> <li>- Variables</li> <li>- IF-THEN-ELSE statements and the Boolean operators AND, OR, XOR, and NOT</li> <li>- FOR and WHILE loops and the integer operators MOD and DIV</li> <li>- ARRAY Variables</li> </ul> </li> <li>4. Demonstrate the ability to combine the above skills to implement algorithms to algorithms to simulate movement and diffusion of biomolecules within and between compartments and the interaction of individual molecules with each other (binding, modification, degradation).</li> <li>5. Demonstrate understanding of biomolecular dynamic equilibria and steady state systems</li> </ol> |   |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>- Computer Practicals including introductory short lectures (20-30 min.)</li> <li>- Individual assignments on computer programming</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <ul style="list-style-type: none"> <li>-Instruction Binders PART 1 and 2 - Java Computer Programming, description and explanation of the basic programming elements and exercises<br/>(The binders are developed by the instructor team. They are free and will be distributed at the start of the course)</li> <li>-Software - NetBeans: a JAVA programming environment<br/>(The software is free and will be distributed at the start of the course)</li> <li>-Laptop (all operating systems are allowed)<br/>(Each student should always bring his/her own laptop)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | <p>Final exercise to be finished at the end of the course and to be discussed individually at the last day of the course in a 15 minutes oral exam with one of the instructors. The final grades are: PASS or FAIL.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Special Information</b>             | <p>The teachers can be reached at a.houtsmuller@erasmusmc.nl or w.vancappellen@erasmusmc.nl</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Location</b>                        | <p>Erasmus MC Rotterdam</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |

| NB1132                                 | Biophysics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3 |
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| <b>Responsible Instructor</b>          | Prof.dr. C. Joo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/10/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 3<br>3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                 | 3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 3<br>4<br>3B<br>4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | In the last few decades, physicists have developed powerful new tools that have allowed us to peer inside the cell at unprecedented levels of detail. The explosion of new biological information these tools have provided has opened up new areas of investigation into biological processes. In this course, students will learn the fundamental physical concepts that underlie biological phenomena, including diffusion, biopolymers, motor proteins, and stochasticity. Students will build on concepts introduced in Physics 1, Biochemistry, Molecular Biology, and Analysis. Students will also learn how to model diffusion and biopolymers using programming. Basic introduction to programming (Python) is included. |   |
| <b>Study Goals</b>                     | At the completion of this course the student will be able to: <ol style="list-style-type: none"> <li>1. Describe the fundamental length and time scales of biological processes at the molecular and cellular levels</li> <li>2. Describe the mechanical property of biopolymers quantitatively</li> <li>3. Describe the laws of motion at low Reynolds number</li> <li>4. Model diffusive motion in one to three dimensions using theories and simulations</li> <li>5. Describe biological processes using Boltzmann distribution</li> <li>6. Build basic Python programs using fundamental programming constructs like variables, conditional logic, looping, and functions</li> </ol>                                          |   |
| <b>Education Method</b>                | - Weekly lectures<br>- Weekly problem sets and discussion<br>- Weekly programming sessions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Books</b>                           | Physical Biology of the Cell (Phillips, Kondev, Theriot)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | Final report: 10%<br>Final exam: 90%<br>--<br>Resit<br>Final report: 10% (the grade from above carries forward)<br>Retake exam: 90%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Studyload/Week</b>                  | 3 ECTS = 84h<br>~10h/week * 4 weeks = 37h (lectures and exercise classes)<br>It leaves ca. ~8h/week study time at home + 15h exam preparation<br><br>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.                                                                                                                                                                                                                                                                                                              |   |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

| NB1140                                 | Physics 1a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. J.W. Zwanikken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Instructor</b>                      | Dr. M.S. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/10/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Required for</b>                    | Physics 1B, Physics 2, Biophysics, Thermodynamics and Transport, Statistical Physics, Physical Biology of the Cell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>        | Analysis 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | The courses Physics 1A and 1B introduce the basic principles of classical physics. Physics 1A covers classical mechanics and an introduction to continuum physics. All the physical principles will be taught with biological examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Study Goals</b>                     | <p>At the completion of this course the student will be able to:</p> <ul style="list-style-type: none"> <li>- Explain the basic laws of classical mechanics, applied to both particles and continuous materials.</li> <li>- Derive and solve equations of motion for common situations in classical physics, including conservative forces, oscillators and ideal fluid flow.</li> <li>- Apply the laws and equations listed above to various real-life situations.</li> <li>- Analyze macroscopic and microscopic biological phenomena quantitatively.</li> <li>- Apply the basic physical concepts learned here to advanced biology and physics topics (Thermodynamics and Transport; Statistical Physics) and biophysics topics (Biophysics; Physical Biology of the Cell; and Nanotechnology).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Method</b>                | <p>Live lectures, demonstrations, and work sessions, that cover the entire course, and online materials that also cover the entire course. A choice can be made to follow the live sessions, or to study the online materials, or to use a combination of both.</p> <p>For the evaluation, two trajectories can be chosen at the beginning of the course:</p> <p>Trajectory I "The training schedule":</p> <ul style="list-style-type: none"> <li>- 6 preparatory quizzes count for 12% of the final grade</li> <li>- 8 homework sets count for 12% of the final grade</li> <li>- the midterm exam counts for 36% of the final grade</li> <li>- the final exam counts for 40% of the final grade</li> </ul> <p>This trajectory helps with the planning of the course, may save the participant a lot of energy figuring out how to prepare for the exam, and allows the participant to get credits for quizzes and homework. It is designed particularly for those who do not feel so confident with physics, or with planning or managing a larger course.</p> <p>Trajectory II "The independent"</p> <ul style="list-style-type: none"> <li>- the midterm exam counts for 47% of the final grade</li> <li>- the final exam counts for 53% of the final grade</li> </ul> <p>This trajectory allows the participants to follow their own schedule, to take part on the quizzes and work sessions, but does not award credits for the quizzes and homework sets. This trajectory is designed for those who feel more confident with physics, those who have the discipline (or energy) to manage their own course, or those who want to practice these important academic skills. Taking part on the work sessions and quizzes is still strongly recommended, but is not obligatory.</p> <p>The choice of trajectory needs to be communicated via Brightspace during the first week of the course, and cannot be changed during the quarter.</p> |   |
| <b>Literature and Study Materials</b>  | <p>Wolfson, Essential University Physics, 3rd edition, including Mastering Physics.</p> <p>NB: You need the FULL version of the book, NOT the 'special TU Delft version' (which is the same book minus the chapters on thermodynamics). The book comes in a 2-volume package and is also used in physics 1B and physics 2. So please buy both volumes 1 and 2, and the Mastering Physics.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | <p>In addition to the book, lecture notes and slides will be provided online.</p> <p>Assessment:</p> <p>Trajectory I "The training schedule":</p> <ul style="list-style-type: none"> <li>- 6 preparatory quizzes count for 12% of the final grade</li> <li>- 8 homework sets count for 12% of the final grade</li> <li>- the midterm exam counts for 36% of the final grade</li> <li>- the final exam counts for 40% of the final grade</li> </ul> <p>Trajectory II "The independent"</p> <ul style="list-style-type: none"> <li>- the midterm exam counts for 47% of the final grade</li> <li>- the final exam counts for 53% of the final grade</li> </ul> <p>The midterm exam covers the first half of the course, and the final exam covers the entire course, with a strong emphasis on the topics of the second half. The retake exam covers the entire course, in proportion to the attention that the topics received during the quarter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Studyload/Week</b>                  | Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Continuing Courses</b>              | Physics 1B and Physics 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| NB1143                                 | Physics 1B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. J.W. Zwanikken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Dr. M.S. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/9/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Period</b>                | 3<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                 | 3<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                     | 3<br>3A<br>3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | In Physics 1B we continue the exploration of classical physics from Physics 1A, and provide an introduction to electromagnetism. The course consists of a theoretical and an exercise part. The exercise part covers masteringphysics and electric circuit simulations, as well as homework assignments that will be graded using peer grading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>Practically the course consists of plenary lectures followed by working sessions in three subgroups supervised by a TA.</p> <p>At the end of this course you will be able to (Learning Objectives, LO):</p> <ol style="list-style-type: none"> <li>1. Explain and use the basic laws of electrostatics to calculate electric forces, fields and potentials.</li> <li>2. Explain and use the laws of magnetism to calculate magnetic forces, fields.</li> <li>3. Apply the laws of electromagnetism to explain and analyze the action of standard circuit elements (resistor, capacitor, and inductor).</li> <li>4. Analyze standard element alternating-current circuits in terms of frequency response and resonances.</li> <li>5. Write clear and detailed solutions to text book problems, such that they can be understood by your peers.</li> </ol> |   |
| <b>Education Method</b>                | <p>Lectures and exercises</p> <p>Activities:</p> <p>Mastering physics: LO1 and LO2</p> <p>Homework: LO1, LO2, LO3, LO4</p> <p>Peer assessment of Homework: LO5</p> <p>Simulations: LO1, LO2, LO3</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | <p>Required:</p> <ul style="list-style-type: none"> <li>- Essential University Physics, Richard Wolfson, Addison-Wesley</li> <li>- Mastering Physics login code, Richard Wolfson, Addison-Wesley</li> </ul> <p>The book comes in a 2-volume package that is also used in Physics 1A and Physics 2. So please buy both volumes 1 and 2, and the Mastering Physics. If you buy the package second-hand, check that the Mastering Physics license is still valid! (It typically is valid for two years).</p>                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | <p>To pass you always need to score at least 5.8 on the final exam, and have a passing grade from mastering physics.</p> <p>The final grade will then be calculated as:</p> <p>Homework: If the best 3 out of 4 HW sets increase the overall grade, they will account for 20% of final grade. Out of this, 15% will be determined by grade on HW solutions handed in, and 5% on the grade for the peer grading performed.</p> <p>Final Exam: 80% (if HW counted)/(100% if HW not counted)</p> <p>Masteringphysics: pass/fail</p> <p>@Retakers</p> <p>If you passed masteringphysics in a previous year, you do not need to re-do this, though your old homework will not count towards the final grade.</p>                                                                                                                                                 |   |
| <b>Studyload/Week</b>                  | Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| NB1150                                 | Lab Course Track A-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3 |
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| <b>Responsible Instructor</b>          | Dr. L. Mezzanotte                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr. J. Pothof                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/5/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | This practical course is the first experimental experience the students of the Nanobiology degree will receive. The main aim of this course is to introduce the students to an experimental environment in which they will learn the essential skills to perform experiments in the area of nanobiology, biochemistry, physics and chemistry. In particular, students will learn how to setup a basic scientific inquiry, keep track of the experimental approach, handle commonly used bio-analytical methods, process and evaluate data and concisely report their findings.                                                                                                                                                                                                                                                                                                                                |   |
| <b>Study Goals</b>                     | <p>At the end of the course the student will be able to:</p> <p>Understand the safety rules in laboratories and know how to follow them;</p> <p>Maintain a clean and organised lab bench;</p> <p>Keep your digital lab journal in an organised fashion and write lab reports clearly;</p> <p>Understand the ethical importance of keeping an accurate and complete lab journal;</p> <p>Master pipetting;</p> <p>Isolate plasmid DNA and perform digestion reaction with restriction enzymes;</p> <p>Perform PCR Gel electrophoresis;</p> <p>Design primers and understand the basics of recombinant DNA technology for cloning;</p> <p>Apply different techniques to determine protein concentration;</p> <p>Purify a protein using Ion-exchange chromatography;</p> <p>Apply a method to measure enzymatic activity;</p> <p>Ion-Exchange Chromatography;</p> <p>Apply tests of significance to datasets.</p> |   |
| <b>Education Method</b>                | This course consists of a series of 2-hour theoretical lectures, introducing the theory underlying the practical work and providing deepening/broadening background information. Followed by a 4-hour supervised session of practical laboratory work. Additionally, data analysis will take approximately 2 hours in class or at home, as well as time for homework and lab journal maintenance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Relations</b>                | <p>There are four Nanobiology 1st year lab courses:</p> <p>NB1150 Lab Practicum A1</p> <p>NB1151 Lab Practicum A2</p> <p>NB1163 Lab Practicum B1</p> <p>NB1164 Lab Practicum B2</p> <p>Students take either A1 and A2, or B1 and B2. Other combinations are not permitted. Students are assigned into group A or group B by the programme based on course capacity. If students from previous years need one more lab course, they need to discuss with the Study Adviseur which course to take.</p>                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Practical Guide</b>                 | Completion of the lab journal of the previous week is mandatory to get access to the practical of the following week. Students with an incomplete lab journal will be denied access to the following practical.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | <p>The final grade is based on a written exam and report on your experiments.</p> <p>In addition, we will monitor the following three items with a fail-pass:</p> <ol style="list-style-type: none"> <li>1. The quality of your lab journal and its completion before every practical</li> <li>2. 100% attendance during the practicals</li> <li>3. 2 out of 2 assignments should be performed correctly and handed in timely</li> </ol> <p>A single fail will result in us not evaluating your exam. In that case, you will need to update your lab journal and/or redo (a) practical(s).</p>                                                                                                                                                                                                                                                                                                                |   |
| <b>Enrolment / Application</b>         | <p>This class is only for students in the Nanobiology BSc programme. Students who wish to take the course will be assigned into this course or its partner course, NB1150. Students who take this course, must take NB1164 as their second lab course to fulfill this requirement.</p> <p>Exchange or bridging students wishing to take a Nanobiology lab course must contact the Nanobiology programme coordinator two weeks before the course starts by emailing <a href="mailto:info-bsc-nb@tudelft.nl">info-bsc-nb@tudelft.nl</a>.</p>                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Studyload/Week</b>                  | <p>3ECTS = 84 hours = 8.4 hours/week</p> <p>Lectures and practicals 6 hrs/week * 7 weeks = 42 hours</p> <p>Homework 2 hrs/week * 8 weeks = 16 hours</p> <p>Exams (midterm and final) 3 hrs/week * 2 weeks = 6 hours</p> <p>Study course material at home 2hrs/week * 10 weeks = 20 hours</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Location</b>                        | Hogeschool Rotterdam and Erasmus MC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |



| NB1151                                 | Lab Course Track A-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. H.J. Geertsema                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Dr. M.E. Aubin-Tam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | This practical course is the first experimental experience the students of the Nanobiology degree will receive. The main aim of this course is to introduce the students to an experimental environment in which they will learn the essential skills to perform experiments in the area of nanobiology, physics and chemistry. In particular, students will learn how to setup a basic scientific inquiry, keep track of the experimental approach, handle commonly used microscopy methods, process and evaluate data and concisely report their findings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                     | <p>At completion of this course, you will be able to:</p> <ol style="list-style-type: none"> <li>1. Describe how you can be more resourceful in the lab</li> <li>2. Perform DNA electrophoresis experiment to verify biological sample</li> <li>3. Build and use a light microscope to visualize biological sample</li> <li>4. Employ Atomic Force Microscopy to study biological sample</li> <li>5. Analyze the data obtained from gel electrophoresis, microscopy and AFM</li> <li>6. Develop and execute an experiment to probe a physical property of your sample of choice</li> <li>7. Concisely and clearly report in a digital lab journal</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                | This course consists of a theoretical lecture, introducing the theory underlying the practical work and providing deepening/broadening background information, a supervised practical session and data analysis. Some homework will be required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Relations</b>                | <p>There are four Nanobiology 1st year lab courses:</p> <p>NB1150 Lab Practicum A1<br/>NB1151 Lab Practicum A2</p> <p>NB1163 Lab Practicum B1<br/>NB1164 Lab Practicum B2</p> <p>Students take either A1 and A2, or B1 and B2. Other combinations are not permitted. Students are assigned into group A or group B by the programme based on course capacity. If students from previous years need one more lab course, they need to discuss with the Study Adviseur which course to take.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Prerequisites</b>                   | Students must have successfully completed NB1150 lab safety and passed lab journal before completing this course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                      | <p>The course grade is calculated by a weighted average of the Exam grade (65%) and the individual assignment (35% total, 5% per assignment).</p> <p>The final exam (2,5 hours) covers all the experimental skills obtained during the course. The individual assignments are to be handed in timely and are graded from 0-10. Deliverable dates for the assignments are communicated through the electronic learning environment.</p> <p>In addition, we will monitor the following two items with a fail-pass:</p> <ol style="list-style-type: none"> <li>1. The quality of your lab journal</li> <li>2. 100% attendance during the practical sessions</li> </ol> <p>Notes:</p> <ul style="list-style-type: none"> <li>- No scores and/or submissions are transferred to the next year.</li> <li>- In order to pass the course, students need to score at least a 5.75 during the final exam</li> <li>- Attendance to all practical sessions is mandatory, exceptions are possible based on a written request with justification (serious illness, family emergencies) to the responsible teacher, Dr. Hylkje Geertsema (h.j.geertsema@tudelft.nl).</li> </ul> |   |
| <b>Enrolment / Application</b>         | Exchange or bridging students wishing to take a Nanobiology lab course must contact the Nanobiology programme coordinator two weeks before the course starts by emailing info-bsc-nb@tudelft.nl.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Studyload/Week</b>                  | <p>3 ECTS = 84 hours = 8.4 hours/week</p> <p>We expect you to spend on:</p> <ul style="list-style-type: none"> <li>- Lectures: 2 hours/week * 10 weeks = 20 hours</li> <li>- Lab work: 4 hours/week * 8 weeks = 32 hours</li> <li>- Homework: 3 hours/week * 10 weeks = 30 hours</li> <li>- Exam prep: 2 hours/week * 1 week = 2 hours</li> </ul> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| NB1163                                 | Lab Course Track B-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. H.J. Geertsema                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. M.E. Aubin-Tam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/5/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | This practical course is the first experimental experience the students of the Nanobiology degree will receive. The main aim of this course is to introduce the students to an experimental environment in which they will learn the essential skills to perform experiments in the area of nanobiology, physics and chemistry. In particular, students will learn how to setup a basic scientific inquiry, keep track of the experimental approach, handle commonly used microscopy methods, process and evaluate data and concisely report their findings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Study Goals</b>                     | <p>At completion of the course, you will be able to:</p> <ol style="list-style-type: none"> <li>1. Apply the safety rules in laboratories</li> <li>2. Conduct basic lab technical skills (i.e. pipetting, preparing buffers, executing a lab protocol)</li> <li>3. Describe how you can be more resourceful in the lab</li> <li>4. Perform DNA electrophoresis experiment to verify biological sample</li> <li>5. Build and use a light microscope to visualize a biological sample</li> <li>6. Employ Atomic Force Microscopy to study a biological sample</li> <li>7. Analyze the data obtained from gel electrophoresis, microscopy and AFM</li> <li>8. Develop and execute an experiment to probe a physical property of your sample of choice</li> <li>9. Concisely and clearly report in a digital lab journal</li> </ol>                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                | This course consists of a theoretical lecture, introducing the theory underlying the practical work and providing deepening/broadening background information, a supervised practical session and data analysis. Some homework will be required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Relations</b>                | <p>There are four Nanobiology 1st year lab courses:</p> <p>NB1150 Lab Practicum A1<br/>NB1151 Lab Practicum A2</p> <p>NB1163 Lab Practicum B1<br/>NB1164 Lab Practicum B2</p> <p>Students take either A1 and A2, or B1 and B2. Other combinations are not permitted. Students are assigned into group A or group B by the programme based on course capacity. If students from previous years need one more lab course, they need to discuss with the Study Adviseur which course to take.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | <p>The course grade is calculated by a weighted average of the Exam grade (65%) and the individual assignment (35%, 5% per assignment).</p> <p>The final exam (2,5 hours) covers all the experimental skills obtained during the course. The individual assignments are to be handed in timely and are graded from 0-10. Deliverable dates for the assignments are communicated through the electronic learning environment.</p> <p>In addition, we will monitor the following two items with a fail-pass:</p> <ol style="list-style-type: none"> <li>1. The quality of your lab journal</li> <li>2. 100% attendance during the practical sessions</li> </ol> <p>Notes:</p> <ul style="list-style-type: none"> <li>- No scores and/or submissions are transferred to the next year.</li> <li>- In order to pass the course, students need to score at least a 5.75 during the final exam</li> <li>- Attendance to all practical sessions is mandatory, exceptions are possible based on a written request with justification (serious illness, family emergencies) to the responsible teacher, Dr. Hylkje Geertsema (h.j.geertsema@tudelft.nl)</li> </ul> |   |
| <b>Enrolment / Application</b>         | <p>This class is only for students in the Nanobiology BSc programme. Students who wish to take the course will be assigned into this course or its partner course, NB1150. Students who take this course, must take NB1164 as their second lab course to fulfill this requirement.</p> <p>Exchange or bridging students wishing to take a Nanobiology lab course must contact the Nanobiology programme coordinator two weeks before the course starts by emailing info-bsc-nb@tudelft.nl.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Studyload/Week</b>                  | <p>3 ECTS = 84 hours = 8.4 hours/week</p> <p>We expect you to spend on:</p> <ul style="list-style-type: none"> <li>- Lectures: 2 hours/week * 10 weeks = 20 hours</li> <li>- Lab work: 4 hours/week * 8 weeks = 32 hours</li> <li>- Homework: 3 hours/week * 10 weeks = 30 hours</li> <li>- Exam prep: 2 hours/week * 1 week = 2 hours</li> </ul> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Location</b>                        | TUDelft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

| NB1164                                 | Lab Course Track B-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3 |
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| <b>Responsible Instructor</b>          | Dr. L. Mezzanotte                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Ing. E. Appeldoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Dr. J. Pothof                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | This practical course is one of the experimental experience the students of the Nanobiology degree will receive. The main aim of this course is to introduce the students to an experimental environment in which they will learn the essential skills to perform experiments in the area of nanobiology, biochemistry, physics and chemistry. In particular, students will learn how to setup a basic scientific inquiry, keep track of the experimental approach, handle commonly used analytical methods, process and evaluate data and concisely report their findings.                                                                                                                                                                                                                                                                                                                 |   |
| <b>Study Goals</b>                     | <p>At the end of the course the student will be able to:</p> <p>Keep your digital lab journal in an organised fashion and write lab reports clearly;</p> <p>Understand the ethical importance of keeping an accurate and complete lab journal;</p> <p>Master pipetting;</p> <p>Isolate plasmid DNA and perform digestion reaction with restriction enzymes;</p> <p>Perform PCR Gel electrophoresis;</p> <p>Design primers and understand the basics of recombinant DNA technology for cloning;</p> <p>Know the basics of bacterial culture and apply optical detection methods to image luminescent bacteria;</p> <p>Apply different techniques to determine protein concentration;</p> <p>Purify a protein using Ion-exchange chromatography;</p> <p>Apply a method to measure enzymatic activity;</p> <p>Ion-Exchange Chromatography;</p> <p>Apply tests of significance to datasets.</p> |   |
| <b>Education Method</b>                | This course consists of a series of 2-hour theoretical lecture, introducing the theory underlying the practical work and providing deepening/broadening background information. Followed by a 4-hour supervised session of practical laboratory work. Additionally, data analysis will take approximately 2 hours in class or at home, as well as time for homework and lab journal maintenance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Relations</b>                | <p>There are four Nanobiology 1st year lab courses:</p> <p>NB1150 Lab Practicum A1</p> <p>NB1151 Lab Practicum A2</p> <p>NB1163 Lab Practicum B1</p> <p>NB1164 Lab Practicum B2</p> <p>Students take either A1 and A2, or B1 and B2. Other combinations are not permitted. Students are assigned into group A or group B by the programme based on course capacity. If students from previous years need one more lab course, they need to discuss with the Study Adviseur which course to take.</p>                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Practical Guide</b>                 | Student must complete the lab journal of previous week to access to practical of the week after. Students who do not complete the weekly lab journal assignment will be denied access to the following practical.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Prerequisites</b>                   | Students must have successfully completed NB1163 and passed lab journal before completing this course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | <p>The final grade is based on a written exam and report on your experiments.</p> <p>In addition, we will monitor the following three items with a fail-pass:</p> <ol style="list-style-type: none"> <li>1. The quality of your lab journal and its timely completion</li> <li>2. 100% attendance during the practicals</li> <li>3. 2 out of 2 assignments should be performed correctly and handed in timely</li> </ol> <p>A single fail will result in us not evaluating your exam. In that case, you will need to update your lab journal and/or redo (a) practical(s).</p>                                                                                                                                                                                                                                                                                                              |   |
| <b>Enrolment / Application</b>         | Exchange or bridging students wishing to take a Nanobiology lab course must contact the Nanobiology programme coordinator two weeks before the course starts by emailing <a href="mailto:info-bsc-nb@tudelft.nl">info-bsc-nb@tudelft.nl</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Studyload/Week</b>                  | <p>3ECTS = 84 hours = 8.4 hours/week</p> <p>Lectures and practicals 6 hrs/week * 7 weeks = 42 hours</p> <p>Homework 2 hrs/week * 8 weeks = 16 hours</p> <p>Exams (midterm and final) 3 hrs/week * 2 weeks = 6 hours</p> <p>Study course material at home 2hrs/week * 10 weeks = 20 hours</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                         |   |

| NB1201                                 | Analysis 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 5 |
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| <b>Responsible Instructor</b>          | Dr. F.J. van de Bult                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 12/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                     | 1<br>2A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Required for</b>                    | <p>This course will teach the basic mathematics which will be relevant to many other courses in the Nanobiology programme.</p> <p>Some examples of courses which rely especially on this course are<br/>           Analysis 2 and 3 (NB1206 and NB1211)<br/>           Physics 1A, 1B and 2 (NB1140, NB1143, NB2141)<br/>           Differential Equations (NB2191)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Expected prior knowledge</b>        | <p>High school mathematics, to be specific the contents of the Dutch Mathematics B.</p> <p>That is:</p> <ul style="list-style-type: none"> <li>* Functions: Polynomials, Power functions, Trigonometric functions, Exponential functions and logarithms.</li> <li>* Algebraic manipulation of formulas</li> <li>* Standard graphs of functions</li> <li>* Differentiation (including product rule, chain rule, quotient rule).</li> <li>* Integration basics</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | <p>In this course you continue the study of functions <math>f(x)</math> of one variable that you started in high school. You will learn a few more techniques for evaluating integrals. You will learn how to find functions that satisfy differential equations such as <math>F = ma</math> (which involves the second derivative <math>a</math> of the location). To do this, it turns out we also have to consider solutions to <math>x^2 = -1</math> (and yes, this equation will have solutions.)</p> <p>You will also study what happens when you add infinitely many (tiny) numbers. One important application of such a sum is that it can be used to find better and better approximations of quantities and functions which cannot be calculated explicitly.</p> <p>Topics of Analysis: complex numbers, linearization, methods of integration, ordinary differential equations, (Taylor) series.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>After this course the student should have achieved the following learning goals:</p> <ol style="list-style-type: none"> <li>1. The student can apply differentiation in various settings:               <ol style="list-style-type: none"> <li>a. Using (inverse) trigonometric and hyperbolic functions,</li> <li>b. For implicitly defined functions,</li> <li>c. To approximate functions and changes in function values</li> <li>d. In applications of the Fundamental Theorem of Calculus</li> </ol> </li> <li>2. The student can apply integration of univariate functions in various settings:               <ol style="list-style-type: none"> <li>a. Calculate integrals using the substitution rule, partial fractions, and integration by parts.</li> <li>b. Determine convergence (and value) of improper integrals</li> </ol> </li> <li>3. The student can work with univariate ordinary differential equations:               <ol style="list-style-type: none"> <li>a. Approximate solutions using Euler's rule</li> <li>b. Solve 1st order linear and separable equations</li> <li>c. Solve 2nd order linear equations with constant coefficients.</li> </ol> </li> <li>4. The student can work with complex numbers               <ol style="list-style-type: none"> <li>a. Do basic calculations</li> <li>b. Work with polar form of complex numbers</li> <li>c. Solve simple equations involving complex numbers (quadratic equations and binomial equations)</li> </ol> </li> <li>5. The student can work with series               <ol style="list-style-type: none"> <li>a. Apply tests to determine convergence of a series ("divergence", integral, comparison, ratio, and alternating series tests)</li> <li>b. Obtain error bounds on the difference of a full series and partial sums</li> </ol> </li> <li>6. The student can work with power series               <ol style="list-style-type: none"> <li>a. Determine interval and radius of convergence.</li> <li>b. Determine Taylor series of a function</li> <li>c. Work with Taylor's theorem on the error of a Taylor polynomial</li> </ol> </li> <li>7. For all the above topics the students can give a few situations (esp. outside mathematics) where it can be applied and is able to apply them to those situations.</li> </ol> |   |
| <b>Education Method</b>                | <p>(Assuming no new COVID measures)</p> <p>This course will include the following educational activities:</p> <ul style="list-style-type: none"> <li>- Interactive lectures</li> <li>- Work sessions where you can practice under the guidance of student assistants</li> <li>- Homework to do at home both on computer and on paper</li> </ul> <p>Apart from basic questions where you can directly apply the methods learned in class, the homework will also contain questions where you have to be more creative. This will help develop your problem solving skills. (The exam can contain such a more creative question or require one of the techniques you had to discover for yourself in one of the homework problems.)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Computer Use</b>                    | <p>Digitally corrected exercises are part of the homework.</p> <p>A digital book is used for the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Literature and Study Materials</b>  | <ul style="list-style-type: none"> <li>- Openstax Calculus volumes 1, 2, and 3.</li> </ul> <p>Download the original books for free from <a href="https://openstax.org">openstax.org</a>, or access them online through Brightspace.</p> <ul style="list-style-type: none"> <li>- There will be some lecture extra notes on complex numbers written by the lecturer</li> </ul> <p>The lecture notes are available on Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                      | <p>(Assuming no new COVID measures)</p> <p>The (regular) assessment consists of</p> <ul style="list-style-type: none"> <li>- Two written or computer exams (2x1.5 hours) [depending on whether it is possible to arrange it, the computer is just for automated grading of an otherwise identical exam]; the exam could even be mixed (part on computer, part written). Both exams have equal weight and there is no minimum requirement.(Grades M and F)</li> <li>- There are bonus quizzes, which count for 20%, but can increase your grade by at most 0.5 points. (Average quiz grade is Q)</li> </ul> <p>The final grade of regular part is thus</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

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|                                         | $\min( (M+F+1)/2, \max( (M+F)/2, (2M+2F+Q)/5 ) )$                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                         | The resit is a single 3 hour exam, for which the quizzes do not count.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                         | You have to register in Osiris for both exams and the resit. You do not have to register for the quizzes. Intermediate grades for the regular exam (M, F or Q) cannot be kept for future years.                                                                                                                                                                                                                                                                                                             |
|                                         | The compensation rule Analysis for Nanobiology is applicable. If the grades for Analysis 1, 2, and 3 (NB1201, NB1206, and NB1211 or prior incarnations) are all at least 5.0 (in decimals), and the weighted (by EC) average of these grades equals at least 5.8 after rounding to decimals, then the student gets the EC for all three courses.                                                                                                                                                            |
| <b>Permitted Materials during Tests</b> | Pen, pencil and non-graphing calculator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Studyload/Week</b>                   | <p>5ECTS = 140 hours total</p> <p>Regular weeks (8 times):<br/>6 hour lecture + 4 hour work session + 4 hour self study = 14 hour/week</p> <p>Exam week (2 times):<br/>12.5 hour exam preparation + 1.5 hour exam = 14 hour/week</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p> |
| <b>Location</b>                         | TU Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| NB1206                                 | Analysis 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5 |
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| <b>Responsible Instructor</b>          | Dr. F.J. van de Bult                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/10/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                     | 2<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Required for</b>                    | This course will teach some of the basic mathematics which will be relevant to many other courses in the Nanobiology programme.<br><br>Some examples of courses which rely especially much on this course are<br>Analysis 3 (NB1211)<br>Physics 1A, 1B, 2 (NB1140, NB1143, NB2141)<br>Signals and Systems (TN2545)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Expected prior knowledge</b>        | This course will assume knowledge of Analysis 1: NB1201 (or earlier incarnation NB1200, WI1415NB, WI1411NB). Especially the parts on integration techniques and series.<br><br>You are also expected to have some limited knowledge about vectors (the part which is taught in Dutch high schools: The basic concept, addition and dot products)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | The main part of this course is about functions of more than 1 variable. To do this we start by reviewing and expanding your knowledge on vectors.<br>Subsequently you will learn how to generalize derivatives and integrals to this new setting. You will also learn some basic applications like finding the maximum of a function of more than 1 variable.<br><br>As a secondary topic we consider Fourier analysis. This is the mathematics behind spectra (for example decomposing music in notes of different frequencies).<br><br>Finally we will spend some time on describing curves (paths through space) and their properties.<br><br>Topics of Analysis: vectors, functions of several variables, partial differentiation, multiple integration, Fourier series and transformation, space curves.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Study Goals</b>                     | After this course the student is expected to have achieved the following learning goals:<br><br>1. The student can calculate with vectors<br>a. evaluate/manipulate expressions involving dot and cross products<br>b. describe lines and planes using vectors<br>2. The student can transform different descriptions of curves (words, equations, parametrisations) into each other and use these to calculate quantitative properties of such curves.<br>a. arc length<br>b. curvature and torsion<br>c. Frenet frame<br>3. The student can apply the theory of differentiation and integration to functions of two variables:<br>a. The student can visualize functions of two variables (using graphs and level curves).<br>b. The student can calculate partial derivatives<br>c. The student can apply the chain rule for functions of several variables<br>d. The student can find the directional derivative and gradient<br>e. The student can find tangent planes and lines to graphs and level curves/surfaces.<br>f. The student can calculate double integrals (including using polar coordinates)<br>g. The student can determine critical points of a function and their type<br>h. The student can obtain extreme values of a function on several domains (both directly and with Lagrange multipliers)<br>4. The student can calculate Fourier series and transformations of simple functions, and apply basic properties of these transformations.<br>a. Obtain the coefficients of Fourier (co)(sine) series<br>b. Describe the convergence of Fourier series (including Gibbs effect)<br>c. Obtain a Fourier transformation of a function<br>d. Evaluate a convolution product (and describe the relation with Fourier transformations)<br>e. Rewrite differential equations to an equation for a Fourier transformation.<br>5. For all the above topics the students can give a few situations (esp. outside mathematics) where it can be applied and is able to apply them to those situations. |   |
| <b>Education Method</b>                | (Assuming no new COVID measures)<br><br>This course will include the following educational activities:<br>- Interactive lectures<br>- Work sessions where you can practice under the guidance of student assistants<br>- Homework to do at home both on computer and on paper<br><br>Apart from basic questions where you can directly apply the methods learned in class, the homework will also contain questions where you have to be more creative. This will help develop your problem solving skills. (The exam can contain such a more creative question or require one of the techniques you had to discover for yourself in one of the homework problems.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Computer Use</b>                    | Digitally corrected exercises are part of the homework.<br><br>A digital book is used for the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | - Openstax Calculus volumes 1, 2, and 3.<br><br>Download the original books for free from openstax.org, or access them online through Brightspace.<br><br>- There will be some lecture extra notes on Fourier and torsion written by the lecturer<br><br>The lecture notes are available on Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | (Assuming no new COVID measures)<br><br>The (regular) assessment consists of<br>- Two written or computer exams (2x1.5 hours) [depending on whether it is possible to arrange it, the computer is just for automated grading of an otherwise identical exam]; the exam could even be mixed (part on computer, part written). Both exams                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>have equal weight and there is no minimum requirement.(Grades M and F)</p> <p>- There are bonus quizzes, which count for 20%, but can increase your grade by at most 0.5 points. (Average quiz grade is Q)</p> <p>The final grade of regular part is thus<br/> <math>\min((M+F+1)/2, \max((M+F)/2, (2M+2F+Q)/5))</math></p> <p>The resit is a single 3 hour exam, for which the quizzes do not count.</p> <p>You have to register in Osiris for both exams and the resit. You do not have to register for the quizzes. Intermediate grades for the regular exam (M, F or Q) cannot be kept for future years.</p> <p>The compensation rule Analysis for Nanobiology is applicable. If the grades for Analysis 1, 2, and 3 (NB1201, NB1206, and NB1211 or prior incarnations) are all at least 5.0 (in decimals), and the weighted (by EC) average of these grades equals at least 5.8 after rounding to decimals, then the student gets the EC for all three courses.</p> |
| <b>Permitted Materials during Tests</b> | Pen, pencil and non-graphing calculator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Studyload/Week</b>                   | <p>5ECTS = 140 hours total</p> <p>Regular weeks (8 times):<br/> 6 hour lecture + 4 hour work session + 4 hour self study = 14 hour/week</p> <p>Exam week (2 times):<br/> 12.5 hour exam preparation + 1.5 hour exam = 14 hour/week</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Location</b>                         | TU Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| NB1211                                  | Analysis 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. F.J. van de Bult                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/6/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                      | 3<br>4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Required for</b>                     | This course will teach some slightly more advanced mathematics which will be relevant to many other courses in the Nanobiology programme.<br><br>Some examples of courses which rely especially on this course are<br>Physics 1B, 2 (NB1143, NB2141)<br>[Unfortunately due to the scheduling of the courses for Physics 1B this mostly means referring back]<br>Thermodynamics and Transport (NB2011)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>         | Analysis 1 and 2: NB1201 + NB1206 (or earlier incarnations: WI1411NB/WI1415NB/NB1200 and WI1412NB/WI1422NB/WI1423NB/NB1205)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                  | In this course you will learn more about integrals and derivatives of functions of 2 and 3 variables. In particular you will learn about different types integrals over curves and surfaces in space. You will also learn about the corresponding fundamental theorems of Calculus (differentiation and integration cancel each other) related to these integrals.<br><br>Vector Calculus: change of coordinates for multiple integrals. Line integrals. Surface integrals. Green, Gauss and Stokes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Study Goals</b>                      | After this course the student is expected to have achieved the following learning goals.<br><br>1. The student can obtain derivatives of implicitly defined functions (also when defined by multiple equations)<br>2. The student can evaluate triple integrals.<br>3. The student can change coordinates in multivariate integrals (more general setting for 2D integrals, to cylindrical and spherical coordinates for 3D integrals).<br>4. The student can calculate the different types of integrals from 2D and 3D vector calculus using parametrizations<br>a. The student can parametrize implicitly defined curves and surfaces.<br>b. The student can evaluate a line integral of a scalar field on $\mathbb{R}^2$ or $\mathbb{R}^3$<br>c. The student can evaluate a line integral of a vector field on $\mathbb{R}^2$ or $\mathbb{R}^3$<br>d. The student can evaluate a surface integral of a scalar field on $\mathbb{R}^3$<br>e. The student can evaluate a flux integral<br>5. The student can apply the relevant version of the Fundamental Theorem of Calculus (=generalized Stokes Theorem) to calculate the integrals of study goal 4<br>a. Fundamental theorem for line integral of a vector field (potentials)<br>b. Green<br>c. Stokes<br>d. Gauss<br>6. The student can perform calculations using gradient, divergence, and curl<br>a. Evaluate them<br>b. Apply product and chain rules<br>c. Determine (vector) potentials<br>d. Work with second order derivatives (such as Laplacian)<br>7. For all the above topics the students can give a few situations (esp. outside mathematics) where it can be applied and is able to apply them to those situations. |   |
| <b>Education Method</b>                 | (Assuming no new COVID measures)<br><br>This course will include the following educational activities:<br>- Interactive lectures<br>- Work sessions where you can practice under the guidance of student assistants<br>- Homework to do at home both on computer and on paper<br><br>Apart from basic questions where you can directly apply the methods learned in class, the homework will also contain questions where you have to be more creative. This will help develop your problem solving skills. (The exam can contain such a more creative question or require one of the techniques you had to discover for yourself in one of the homework problems.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Computer Use</b>                     | Digitally corrected exercises are part of the homework.<br><br>A digital book is used for the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature and Study Materials</b>   | - Openstax Calculus volumes 1, 2, and 3.<br><br>Download the original books for free from openstax.org, or access them online through Brightspace.<br><br>- There will be some lecture extra notes on implicit differentiation written by the lecturer<br><br>The lecture notes are available on Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Assessment</b>                       | (Assuming no new COVID measures)<br><br>The (regular) assessment consists of<br>- one written or computer exams (2 hours) [depending on whether it is possible to arrange it, the computer is just for automated grading of an otherwise identical exam]; the exam could even be mixed (part on computer, part written). (Grade E)<br>- There is a bonus midterm, which count for 20%, but can increase your grade by at most 0.5 points. (Midterm grade is M)<br><br>The final grade of regular part is thus<br>$\min((2E+1)/2, \max(E, (4E+M)/5))$<br><br>The resit is a single 2 hour exam, for which the midterm does not count.<br><br>You have to register in Osiris for midterm, exam and resit. Intermediate grades for the regular exam (E or M) cannot be kept for future years.<br><br>The compensation rule Analysis for Nanobiology is applicable. If the grades for Analysis 1, 2, and 3 (NB1201, NB1206, and NB1211 or prior incarnations) are all at least 5.0 (in decimals), and the weighted (by EC) average of these grades equals at least 5.8 after rounding to decimals, then the student gets the EC for all three courses.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Permitted Materials during Tests</b> | Pen, pencil, and non-graphing calculator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |



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|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Studyload/Week</b> | <p>3ECTS = 84 hours</p> <p>Regular weeks (8 times)<br/>4 hour lecture + 2 hour work session + 2 hour self study = 8 hour/week</p> <p>Midterm (once)<br/>7 hour preparation + 1 hour midterm = 8 hour/week</p> <p>Final (once)<br/>10 hour preparation + 2 hour final = 12 hour/week</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p> |
| <b>Location</b>       | TU Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| NB1230                                  | Linear Algebra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3 |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. I.M. Smit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Required for</b>                     | Linear Algebra is an important mathematical skill and language that will be used throughout the program. In particular, Linear Algebra plays a significant role in the courses on Differential Equations (NB2191), Quantum Mechanics (NB3017,NB3018), Systems and Signals (TN2545) and Image Analysis (NB2121).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>         | <ul style="list-style-type: none"> <li>- High school mathematics B, or equivalent.</li> <li>- Analysis 2 (NB1205) (the part on vectors).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                  | <p>Linear Algebra is a branch of mathematics which is based on two mathematical operations: addition and (scalar) multiplication. Despite this deceptive simplicity, it has a vast variety of applications, both within and outside mathematics. In the Nanobiology program, Linear Algebra plays a role in several courses, such as Analysis, Differential Equations, Quantum Mechanics, Systems and Signals and Image Analysis.</p> <p>This course is an introduction to several fundamental aspects of Linear Algebra, like vector calculation, solving systems of linear equations and matrix algebra, and elementary notions like linear combination, linear dependence, basis, dimension, eigenvalues and eigenvectors, standard inner product and orthogonal projections.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Study Goals</b>                      | <p>The most important learning objectives for this course are given below.</p> <p>The student</p> <ol style="list-style-type: none"> <li>1. can solve systems of linear equations and make a priori statements about the number of solutions;</li> <li>2. can perform matrix operations (addition, multiplication, inverse, determinant) and can determine the matrix of a linear transformation;</li> <li>3. can determine whether a set of vectors is linearly dependent or linearly independent, and relate this to other concepts in linear algebra;</li> <li>4. can determine dimension and bases of linear subspaces, in particular of the null space and column space of a matrix;</li> <li>5. can find the (possibly complex) eigenspaces and eigenvalues of a given matrix;</li> <li>6. can determine whether a matrix is diagonalisable or not, can diagonalise a matrix if possible and use this to analyse dynamical systems;</li> <li>7. can find orthogonal decomposition of a vector with respect to a linear subspace;</li> <li>8. can find least square solutions of a system of linear equations, and use this to fit curves to data.</li> </ol> <p>Please note: the student should know and understand all terms mentioned, as well as the notions that are implicitly needed to comply with the learning objectives.</p> |   |
| <b>Education Method</b>                 | <ul style="list-style-type: none"> <li>- In a general week, there are four hours of interactive lectures ("colstructies"), in which the teacher presents new material. There will also be time to work on and discuss exercises.</li> <li>- In addition, there are 2 hours of tutorial classes, in which there is time to work on homework under supervision of the teacher and teaching assistants. These hours will also be used to discuss some of the more challenging exercises.</li> <li>- There are two bonus quizzes.</li> <li>- Homework consists of computation-oriented exercises in GraspLe and more conceptual exercises provided on Brightspace in pdf format.</li> </ul> <p>Your education in linear algebra will help you build your professional skills in problem solving, writing (in the language of linear algebra) and team work. Even though your exam is individual, cooperating during exercise class will help you learn the material more thoroughly.</p> <p>* Due to the ongoing covid pandemic, schedules might be slightly different.</p>                                                                                                                                                                                                                                                                      |   |
| <b>Literature and Study Materials</b>   | <ul style="list-style-type: none"> <li>- Book: Lay, Linear Algebra and Its Applications, Global Edition, 6th edition, ISBN 978-1292351216. Using an older edition of this book is no problem. This book can be obtained through the student association or StudyStore (<a href="http://www.studystore.nl">www.studystore.nl</a>).</li> <li>- GraspLe: An online homework environment that you can access through Brightspace.</li> <li>- Problem sets on PDF, available on Brightspace.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                       | <p>The assessment consists of:</p> <ul style="list-style-type: none"> <li>- Two bonus quizzes (~30 minutes each) for in-between feedback on your progress.</li> <li>- A final 2-hour exam.</li> </ul> <p>The final grade equals the exam grade, but the quizzes can give you a bonus: if the average quiz grade is higher than the exam grade, then the exam grade is increased by 20% of the grade difference, with a maximum of 0.5 point. In formula:<br/> <math display="block">F = E + \min\{0.5, \max\{0, 0.2(Q - E)\}\}</math> where F is the final grade, E the exam grade and Q the average quiz grade. You passed the course whenever the final grade is at least 5.75.</p> <p>The resit consists of a 2-hour exam. Note that if you take the resit exam, the aforementioned quizzes will not contribute to your grade.</p> <p>Registration for exams is done through Osiris.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Permitted Materials during Tests</b> | Pen and paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Studyload/Week</b>                   | <p>The total study load is 3 ECTS = 84 hours.</p> <p>I would suggest to distribute these as follows:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |

- 16 x 2 equals 32 hours of lectures.
- 8 x 2 = 16 hours of exercise class.
- 8 x 2.5 = 20 hours on homework
- 1 hour quiz time
- 13 hours exam preparation
- 2 hours exam time

Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.

**Location**

Delft

|                     |                             |
|---------------------|-----------------------------|
| <b>Year</b>         | <b>2022/2023</b>            |
| <b>Organization</b> | <b>Applied Sciences</b>     |
| <b>Education</b>    | <b>Bachelor Nanobiology</b> |

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|-----------------------------------------|---------------|
| <b>2e jaar BSc NB 2022</b>              |               |
| <b>Responsible Program<br/>Employee</b> | J.C. Colgrove |

| NB2011                                 | Thermodynamics and Transport                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | P. Boukany                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                      | Dr. B. Bera                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>This course deals with transport of mass (or moles), and energy - the basic quantities in classical physics - at the level of cell. Most processes involve transport- or transfer in general - of these quantities. Mass, fluid and heat transport processes are fundamental to virtually all aspect of life. For example, living systems need energy to survive. Moreover, they need new material, in the form of specific molecules to grow and reproduce or get their energy from.</p> <p>Thermodynamics:<br/>           -First law of thermodynamics; Equation of States; Changing States<br/>           -Entropy and second law of thermodynamics; Reversible process and work/heat;<br/>           -Entropy, Enthalpy and Gibbs Free Energy; Chemical potential</p> <p>Transport phenomena:<br/>           -Microscopic theory of Diffusion; Brownian motion; Random Walks<br/>           -Macroscopic description of diffusion; Fick's equation; Diffusion through pores<br/>           -Mass transport across a thin film; Reaction-Diffusion system;<br/>           -Life at low Reynolds number; Diffusion with drift; Stokes flow; Flow through a channel; Sedimentation flow</p> <p>Mathematical analysis methods: scaling and approximation techniques, analytical and numerical approaches.</p> |   |
| <b>Study Goals</b>                     | <p>After this course the student should:</p> <ul style="list-style-type: none"> <li>-have knowledge of the basic definitions and concepts of thermodynamics and transport phenomena at the scale of the cell;</li> <li>-be able to analyse basic problems concerning heat, mass and momentum transfer in and around (living) cells;</li> <li>-be able to model the related processes and solve the models to obtain a quantitative description of the situation; be able to assess the outcome of the analysis using a first principle approach.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | <p>Class room lectures followed by exercises and guided self study.<br/>           Note: The exact format of course may be changed due to Coronavirus (COVID-19) regulations.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Books</b>                           | Random walks in biology, Howard C. Berg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Prerequisites</b>                   | <p>BSc: Physics 1<br/>           Note: it is absolutely required to have a good working understanding of BSc level Mathematics.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | <p>Mid term exam (Thermo) and final exam (Transport): both are of the problem solving type.<br/>           In addition, during the course two assignments will be handed out. First assignment is related to thermodynamics, and includes the results from a practicum. The second assignment is related to transport phenomena. These assignments are graded; the grading forms part of the final mark.</p> <p>Final mark = (10%) assignment-1 (Thermo) + (10%) assignment-2 (Trans) + (40%) midterm exam (Thermo) + (40%) final exam (Trans)</p> <p>The grades for the assignment are only valid for this academic year. That means you can keep the assignment grades for the resit, but not for next year. In case you have to resit, only the exams can be resit, and only a combined resit exam covering both topics can be taken.</p> <p>Note: The exact format of assessment may be changed due to Coronavirus (COVID-19) regulations.</p>                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Studyload/Week</b>                  | <p>6h/week*8week=48h (lectures+working class)</p> <p>3h/week study (3*8 week=24 hr) time at home 12 h exam preparation</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

| NB2022                                 | Philosophy and Ethics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3 |
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| <b>Responsible Instructor</b>          | M.M. Jaegle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | 2<br>2B<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>In this course the student will be introduced to some key thinkers in the philosophy of science and will learn the fundamental philosophical underpinnings of experimental science. Aspects of scientific integrity and ethics will be discussed.</p> <p>The understanding gained in this course is needed to appreciate how the knowledge learned in other courses such as molecular biology and biophysics was generated and to appreciate that this knowledge may change. Explicit discussion of scientific and professional ethics develops a students understanding of their contribution to a cooperative and productive scientific community.</p> <p>Literature and study material:<br/>* Reading materials provided by the lecturers</p>                                                                                                                                  |   |
| <b>Study Goals</b>                     | <p>At the completion of this course the student will be able:</p> <ul style="list-style-type: none"> <li>* to demonstrate knowledge and understanding of some of the key concepts, arguments and positions in the philosophy of science;</li> <li>* to reflect on the power, and also the limitations, of scientific methods of thinking;</li> <li>* to demonstrate knowledge of proper ethical behavior in research;</li> <li>* to get acquainted to biosecurity considerations relevant for doing research;</li> <li>* to demonstrate knowledge and understanding of important ethical values and principles associated with nanobiology research;</li> <li>* to explain what the debate on human enhancement is about, and to describe the two main ethical positions in this debate;</li> <li>* to reflect on ethical issues that arise in biotechnological research;</li> </ul> |   |
| <b>Education Method</b>                | <p>Education methods:<br/>* Lectures and discussion sessions.</p> <p>* Group assignments and individual assignment</p> <p>Professional skills:</p> <ul style="list-style-type: none"> <li>- Group Presentation</li> <li>- Team work</li> <li>- Writing report</li> <li>- Strong Work Ethic</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <ul style="list-style-type: none"> <li>* Reading materials provided by the lecturers</li> <li>* Online lectures provided by the lecturer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Books</b>                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | <p>25% Group assignment "Interview with a scientist": grade</p> <p>25% Group assignment "Presentation on an ethical issue": Pass/fail</p> <p>20% Written exam: grade</p> <p>30% Individual assignment: grade</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Special Information</b>             | This course will be taught all of Q2 beginning in 2022-2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Remarks</b>                         | Martine Jaegle can be reached at m.jaegle@erasmusmc.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Studyload/Week</b>                  | <p>7.5h/week (lectures and assignments)</p> <p>16h for exam preparation</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Location</b>                        | Erasmus MC, teacher can be reached at m.jaegle@erasmusmc.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| NB2032                                 | Evolutionary Developmental Biology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 6 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Ir. H. Mira Bontenbal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                      | Dr.ir. W.M. Baarends                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/8/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 3<br>3A<br>3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                     | 3<br>3B<br>4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | <p>All animal species develop from a single cell, the fertilized egg, and share genes which regulate development. A large number of genes encoding components of molecular and cellular mechanisms involved in development are highly conserved among animal species. Yet, nature shows us an astonishing variety of animal forms. The basis for this variety lies in genome diversification through mechanisms such as gene duplication and mutation events, which generate opportunities for further modification and refinement of developmental schemes, resulting in new forms and functions, if the new phenotypes are successful under natural selection.</p> <p>Evolution and diversification of animal forms occurs through changes in development. By studying developmental biology in an evolutionary context, we gain a much better understanding of the mechanisms at work in animal development. In this course, the students will learn about general aspects of development across the animal kingdom: from worms and flies to fish, birds, and mammals. It goes without saying that this approach also provides an illuminating picture of our own development.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Study Goals</b>                     | <p>Upon completion of this course, the student will be able to:</p> <ul style="list-style-type: none"> <li>- Describe evolutionary conserved concepts of animal development and cell differentiation in various model species (Understand)</li> <li>- Identify embryo stages and key structures at different phases in different model organisms (Remember)</li> <li>- Explain similarities and differences between different model species at the level of gene regulation, cells, embryo structures, organs and limbs; using knowledge on the biological processes that are at the base of those similarities and differences. (Understand)</li> <li>- Describe the basic evolutionary concepts about the development of multicellularity, sexual reproduction, and Darwins theories concerning speciation (Understand)</li> <li>- Connect specific evolutionary events to the appropriate evolutionary concept. (Apply)</li> <li>- Provide the definitions of toti- and pluripotency of cells (Remember)</li> <li>- Assess the possibilities of the use of different types of (induced) pluri- and totipotent stem cells in research and therapeutic applications (Evaluate)</li> <li>- Critically interpret the results of scientific experiments in developmental biology, using provided text and figures from scientific manuscripts. (Analyse)</li> <li>- Develop a research question using information gained from a scientific literature search (Create)</li> <li>- Assess the literature on the poster topic and summarize the findings in components of a poster (Evaluate)</li> <li>- Constructively collaborate with fellow students to prepare a scientific poster, and presentation (not applicable, it is an attitude type of learning aim)</li> <li>- Present and defend a scientific poster to peers in an engaging and comprehensive manner (not applicable, it is an attitude type of learning aim)</li> <li>- Critically evaluate scientific presentations of peers using an assessment rubric (Evaluate)</li> </ul> |   |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>- interactive lectures</li> <li>- interactive practicals</li> <li>- integrative sessions where non-graded homework are corrected and non-graded quizzes are done by the students</li> <li>- homework (non graded)</li> <li>- workshops, meeting with <i>C. elegans</i> and zebra fish</li> <li>- visit to the Natural History Museum Rotterdam</li> <li>- assignment: make a poster (shared responsibility of 2-4 students); these posters will be presented by the students at a poster session, whereby the teachers and the students will assess the quality of each poster using an evaluation form. This will develop your soft skills: group work, poster presentation and peer review.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <ul style="list-style-type: none"> <li>- Principles of Development; Wolpert and Tickle; Sixth Edition 2019</li> <li>- Students should already have: Molecular Biology of the Cell; Alberts et al.; Sixth Edition 2015</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | <p>The final grade consists of three parts</p> <ul style="list-style-type: none"> <li>- Part 1: Midterm exam (open questions and multiple choice questions), passing grade is 5.8 or higher. Electronic exam. This will cover the 1st part of the course.</li> <li>- Part 2: Final exam (open questions and multiple choice questions), passing grade is 5.8 or higher. Electronic exam. This will only cover the 2nd part of the course, not the full course.</li> <li>- Part 3: Poster Presentation, passing grade is 5.8 or higher.</li> </ul> <p>The global course grade is an average of each parts grade (1/3 Part 1, 1/3 Part 2, 1/3 Part 3).</p> <p>Parts 1 and 2 can compensate for each other but Part 3 cannot compensate. This means the student will have to pass the exam (average of Part 1 and Part 2) and Part 3.</p> <p>The grade for Part 3 will be kept until the exam has been passed.</p> <p>The ECTS for this course will only be assigned to you when the online course "Information Literacy 2" is successfully completed.</p> <p>The exams will take place electronically, while the poster session will be physical.</p> <p>In terms of resits:</p> <ul style="list-style-type: none"> <li>- At the exam resit, the student can chose to do Part 1 or Part 2, or both. The highest grade of each Part will be kept for the global course grade.</li> <li>- There is no resit for Part 3.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Special Information</b>             | This course will be taught in Q3 beginning in 2022-2023.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Remarks</b>                         | <p>Hegias Mira Bontenbal can be reached at h.mirabontenbal@erasmusmc.nl</p> <p>Willy Baarends at w.baarends@erasmusmc.nl</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Studyload/Week</b>                  | <p>6ECTS, 2 octals,<br/>6ECTS = 168h</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <p>7h/week * 8 weeks = 56h (lectures and working classes)<br/>Leaves ca. 10h/week study time at home, homework, poster preparation + 32h exam preparation</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p> |
| <b>Schedule</b> | We will put our up-to-date schedule on Brightspace. So please follow that schedule and not the schedule on mytimetable.tudelft.nl. Use the Brightspace one when the course starts.                                                                                                                                                                                                                                                   |
| <b>Location</b> | Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                            |

| <b>NB2041</b>                          | <b>Optics and Microscopy</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>3</b> |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Responsible Instructor</b>          | Dr. F. Bociort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
| <b>Responsible Instructor</b>          | Dr. E.C.M. Carroll                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/8/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Education Period</b>                | 3<br>3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>Start Education</b>                 | 3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Exam Period</b>                     | 3<br>4<br>3B<br>4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>Expected prior knowledge</b>        | You have already encountered some of the topics below at an earlier physics course, this helps you to follow this course better.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>Course Contents</b>                 | <p>Part 1 Optics :</p> <p>Week 1:<br/>Lectures 1.1 and 1.2: Introduction, Electromagnetic waves, Polarization<br/>Lectures 1.3 and 1.4: Interference, Diffraction, Resolution limit</p> <p>Week 2:<br/>Lectures 2.1 and 2.2: Geometrical optics, Imaging, Paraxial approximation, Thin lenses<br/>Lectures 2.3 and 2.4: Visual optical systems, Simple microscope, Matrix formalism for imaging</p> <p>Part 2 Microscopy:</p> <p>Week 3:<br/>Lectures 3.1 and 3.2: Bright field and dark field microscopy and phase contrast microscopy</p> <p>Week 4:<br/>Lectures 4.1 and 4.2: Polarisation microscopy<br/>Lectures 4.3 and 4.4: Fluorescence microscopy and Confocal scanning microscopy</p>                                                |          |
| <b>Study Goals</b>                     | <p>Part 1 : The student will have a good understanding of the basics of optics, including imaging and the knowledge necessary for understanding the construction of a traditional microscope. The student will be able to apply the theory to solve simple problems.</p> <p>Part 2: The student will have a good understanding of several microscope types, including the general optical design and the interpretation of images. The types of microscopes are: bright and dark field microscopes, phase contrast microscopes, polarisation microscopes, fluorescence microscopes and confocal laser scanning microscopes.</p>                                                                                                                |          |
| <b>Education Method</b>                | Lectures and exercises                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>Literature and Study Materials</b>  | <p>Part 1: For all lectures there will be hand-outs of the lecture material. The hand-outs and the exercises (with solutions) are in principle sufficient for preparing the exam. Students who are interested in a deeper understanding of the topics can read E. Hecht, Optics, any edition</p> <p>Part 2: For part 2 the student will have lecture notes and exercises available on Brightspace but some chapters of the book below will be required for further reading:<br/>D. B. Murphy, Fundamentals of Light Microscopy and Electronic Images, J. Wiley&amp;Sons.<br/>This book can be downloaded from the library (per chapter).<br/>The exam will be based on the lecture notes, exercises and the specific chapters of the book.</p> |          |
| <b>Assessment</b>                      | Written examination, both for the main exam and for the resit. Normally the exam is closed-book, off-campus exams are open-book.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>Studyload/Week</b>                  | Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |



| NB2046                          | Microscopy / Nanoscopy practice                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1.5 |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Responsible Instructor          | Prof.dr. A.B. Houtsmuller                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| Responsible Instructor          | Dr. W.A. van Cappellen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| Contact Hours / Week<br>x/x/x/x | 0/0/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| Education Period                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| Start Education                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| Exam Period                     | 3<br>4<br>4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| Course Contents                 | In the practical course Microscopy/Nanoscopy (MiNa) the students will be guided through the most important, up-to-date fluorescence microscopy techniques for quantitatively studying cellular function and dysfunction at the nanoscopic scale. The students will work (in groups of five) on various topics (four will be chosen from a total of six) using different advanced fluorescence bioimaging techniques: cell division (mitosis and meiosis) will be studied using confocal time lapse microscopy and single molecule localisation (SML) (topics 1, 2), the role of focal adhesions in cell migration will be studied using total internal reflection (TIRF) microscopy and structure illumination (super resolution) microscopy (SIM) (topics 3, 4), gene transcription regulation will be studied by fluorescence recovery after photobleaching (FRAP) (topic 5), and cell membranes will be visualised and cell volume/surface ratios will be determined using specific membrane and nuclear staining and 3-D confocal microscopy (topic 6). For each topic the students will use the various state-of-the-art research microscopes to record images themselves (under supervision), will analyse the data, will write a report and in a final session will discuss the reports with the supervisor in a 15 minute session. Participation in all practicals, the introductory lectures and the final presentation session is compulsory.<br>The course is closely related to all Nanobiology courses as a variety of cell biological model systems are investigated using microscopes, analysed using mathematical techniques and reported in writing and discussed afterwards. |     |
| Study Goals                     | At the completion of this course the student will be able to:<br><br>A. demonstrate understanding of the basic principles of:<br>1. fluorescence and green fluorescent protein (GFP)-tagging,<br>2. wide field microscopy, confocal microscopy, TIRF microscopy, and structured illumination microscopy,<br>3. quantification of fluorescent signals in microscope images,<br>4. fluorescence recovery after photobleaching (FRAP) experiments,<br><br>B. write a well-structured report, including discussion, on a (set of) microscopic experiment(s),<br><br>C. demonstrate understanding of the content of the report in a 15 minutes session with a supervisor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| Education Method                | - Introductory Lecture on Microscopy (all students)<br>- Practicals (in groups of 5)<br>- Supervised writing of practical report (one per group on four different topics)<br>- Discussion (in groups of 5) with a supervisor about the reports (15 minutes at the last day of the course)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| Literature and Study Materials  | - Binders and manuals with detailed description of microscopes and microscope assays and detailed background information on the model systems studied<br>- FIJI image analysis software (will be distributed at the start of the course)<br>- The students' own laptop (any operating system is allowed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| Assessment                      | Assessment will be based on individual effort and participation during the practicals, and on the participation in and contribution to<br>1. writing the groups practical reports,<br>2. discussion of the results on the basis of the practical reports (with the supervisor).<br><br>Grading is pass or fail.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| Enrolment / Application         | Students must enroll in the Brightspace page for this course at least one week before the first class session so that they can be grouped.<br>Exchange or bridging students wishing to take this course should contact the programme coordinator at info-bsc-nb@tudelft.nl at least 3 weeks prior to the start of the course to see if there is space available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| Special Information             | The teachers can be reached at a.houtsmuller@erasmusmc.nl or w.vancappelen@erasmusmc.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| Remarks                         | Presence at this course is mandatory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| Location                        | Erasmus MC Rotterdam, Erasmus Optical Imaging Centre                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |

| NB2071                                 | Physical Biology of the Cell 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | D.H.M. Meijer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | Prof.dr. G.H. Koenderink                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Required for</b>                    | This course is required for understanding the cellular basis of development (course Evolutionary & Developmental Biology).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | The course will assume knowledge of cellular biomolecules gained in the courses of Chemistry, Molecular Biology and Genetics. It partly builds on concepts introduced in Physical Biology of the Cell 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>Biological cells are complex biochemical systems that obey the laws of physics. Modern research in molecular and cell biology therefore increasingly relies on physical concepts. In Physical Biology of the Cell 1 you learned how to utilise such concepts to model cellular phenotypes on a macroscopic scale. In Physical Biology of the Cell 2 you will transfer this approach to the molecular scale and learn how signalling cascades, and (experimental) models to interpret them, are essential as we build our molecular understanding of cellular phenomena such as information processing, translocation across membranes, and mechanobiological signalling.</p> <p>In this course you will learn how to connect concepts from biology and biophysics with observations from biochemistry and physics to describe this diversity of cellular processes and to understand their molecular basis. You will then apply these concepts to concrete examples in which you will establish quantitative models for the regulation of biochemical cascades and circuits, the functional dynamics of molecular machines and processes related to translocation and the propagation and processing of biological signals.</p> |   |
| <b>Study Goals</b>                     | <p>Upon completion of this course you will be able to:</p> <ol style="list-style-type: none"> <li>1. Identify fundamental challenges in cellular information processing</li> <li>2. Describe how protein-protein interactions are used to transmit information</li> <li>3. Understand how canonical classes of signaling proteins couple conformational change with catalytic activity</li> <li>4. Understand how changes in post-translational modification and/or subcellular localization of signaling molecules transmit information</li> <li>5. Distinguish functional differences in speed, efficiency and specificity of second messengers</li> <li>6. Understand structural elements and organization of the cell membrane</li> <li>7. Explain the function of mechanobiological signaling in different tissues</li> <li>8. Apply concepts of cell signaling to model signaling pathways</li> <li>9. Interpret schematic information from a figure to both a biophysical and a biological meaning.</li> <li>10. Relate current open questions in cellular information processing to experimental approaches</li> <li>11. Evaluate primary literature on novel technologies to study cellular communication</li> </ol>      |   |
| <b>Education Method</b>                | Lectures (plenary)<br>Assignments (in subgroups)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Literature and Study Materials</b>  | <p>Required textbooks:</p> <ol style="list-style-type: none"> <li>1. Cell Signaling. Principles and mechanisms, Lim, Mayer, Pawson</li> <li>2. Molecular Biology of THE CELL, Alberts</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | <p>Students will be assessed according to the study goals as formulated above. The exam consists of open questions probing your quantitative and qualitative understanding of the study goals.</p> <p>To pass the course you need a 5.8 for your final exam.</p> <p>Grading<br/>Final exam (90% of the final grade, individual)<br/>Homework assignments (10% of the final grade, individual)<br/>Homework assignments will be assessed on a pass/fail scale.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Special Information</b>             | This course will be taught in Q2 beginning in 2022-2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Studyload/Week</b>                  | <p>3 ECTS, octal course = 84 h<br/>5h/week * 8 weeks = 40h (lectures and working classes)<br/>Leaves ca. 10h/week study time at home + exam preparation</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| NB2081                                 | Nanotechnology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. M.E. Aubin-Tam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | Prior to the beginning of this course students are expected to have completed the first year lab courses, and the courses Physics 1A (NB1140), Physics 1B (NB1143), Biophysics (NB1132), and Physics 2 (NB2141).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | In this course the student are introduced to current and developing technology used in Nanobiology research. They will learn the basic principles of instruments such as optical and magnetic tweezers and atomic force microscopy. The use of nanofabrication, microfabrication and microfluidics in nanobiology will be illustrated. The type of data produced, and the relevant sensitivity and accuracy of the different instruments will be described. Students will work hands-on with one of these instruments. The relevant application of specific instruments to answering questions in biology will be discussed. Bottom-up and top-down nanofabrication approaches will be described. This course will assume knowledge of Physics, Math, Biophysics, and Molecular Biology as learned in YR1 and previous YR2 courses. Knowledge learned in this course is needed to understand current advances in nanobiology and make informed choices for thesis project work. |   |
| <b>Study Goals</b>                     | <p>At the completion of this course the student will be able to:</p> <ul style="list-style-type: none"> <li>- Accurately describe the working principle of several (at least 3) instruments currently used in nanobiology research</li> <li>- Identify the type of data each instrument can produce and name relevant applications in nanobiology research</li> <li>- Describe how one such instrument is calibrated</li> <li>- Identify areas of application in medicine that have benefited from nanotechnology instrumentation</li> <li>- Describe bottom-up and top-down nanofabrication approaches.</li> </ul>                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Method</b>                | <p>Interactive lectures<br/> Self / group study practicum and report writing. This will develop these skills: taking safety measures when doing practical work, scientific writing, team work.<br/> Practical work with selected instrument</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | <p>Report(s) and assignments: 50% ; Final exam: 50% ;<br/> Note that the retake exam only allows to change the grade of the final exam, while the grade of the reports/assignments cannot be changed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Studyload/Week</b>                  | <p>2EC = 56h;<br/> 8 lectures of 2h = 16h;<br/> 2 practicum = 28h (8h of practical work, 20h for data analysis and report writing);<br/> 12h for exam preparation.<br/> Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| NB2121                                 | Image Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. D. Brinks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Dr. W.A. van Cappellen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr. E.C.M. Carroll                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Instructor</b>                      | Dr. I.V. Smal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. M.P. Chien                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 4<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Expected prior knowledge</b>        | <p>This course builds on previous courses: Analysis and Linear Algebra (Year 1), and Signals and Systems, Optics and Microscopy, and Computational Science (Year 2).</p> <p>NB2041 Optics and Microscopy, NB2046 Microscopy-Nanoscscopy Practice and this course together form the imaging backbone of the nanobiology curriculum, which can be expanded upon by following the third year elective NB3021 Optics and Applications in Nanobiology.</p> <p>The positioning of this course in the year, after NB2041 and NB2046, is based on the reasoning that first learning the basics of microscopy (NB2041), followed by learning how to make an image on a microscope in practice (NB2046), followed by learning how to analyse that image (this course), is a logical progression. It is unfortunately currently scheduling-wise not possible to run this course in parallel or reverse order with NB2046, for those that would have liked to apply what they learned here on their own images. However, the first practical of this course serves as a brief introduction to ImageJ, which may be handy for NB2046.</p>                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>Microscopy images contain a wealth of information. Extracting this information by visual inspection or manual measurement is not only cumbersome but also subjective. Automation of image analysis tasks by using a computer and the appropriate methods and tools, allows for higher efficiency, accuracy, objectivity, reproducibility, and completeness. In this course the student will learn how to quantitatively analyze microscopy images by computer and will develop the skills to use computational image analysis tools for biological research. The course consists of lecture videos and contact hours, providing the theoretical underpinning of image analysis methods, and practical sessions, in which the student applies learned principles to real microscopy image data and analysis problems using the open-source Java-based bioimage analysis tool ImageJ (National Institutes of Health).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Study Goals</b>                     | <p>Upon successful completion of this course the student will be able to:</p> <ul style="list-style-type: none"> <li>* List the main properties of digital microscope images relevant for quantitative analysis.</li> <li>* Recommend image analysis procedures in either spatial or frequency domain image representation.</li> <li>* Construct image analysis pipelines consisting of linear and nonlinear image processing steps.</li> <li>* Select the appropriate filters for image enhancement and feature detection.</li> <li>* Select the appropriate filters for morphological image processing and analysis.</li> <li>* Describe the strengths and weaknesses of image segmentation approaches.</li> <li>* Compute quantitative properties and measures of image structures.</li> <li>* Describe the main components of image registration algorithms.</li> <li>* Understand the main principles of object tracking and motion analysis.</li> <li>* Explain how volume and surface rendering algorithms work.</li> <li>* list common machine learning techniques used in image analysis and discuss their pros and cons.</li> <li>* Use relevant image processing and analysis tools within ImageJ.</li> <li>* Write image processing and analysis pipelines as scripts in ImageJ.</li> </ul>                                                                                                                                                                                           |   |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>* Lecture videos</li> <li>* contact hours</li> <li>* Practicals</li> </ul> <p>In principle the course will be done in a flipped format. The student is supposed to start every weekly block by watching lecture videos of the topic of the week. This is followed by contact hours with the lecturers.</p> <p>The content of the contact hours and lecture videos will partly be determined by feedback from the students (which we only obtain after this study guide text needs to be finalized) and is subject to change; however, there will always be opportunity to ask questions and obtain clarification on the lecture videos of the week. The contact hours are followed by practicals where the students learn how to use the concepts they learned this week in ImageJ with the help of TAs.</p> <p>Contact hours and practicals will be either live or online, depending on the covid situation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Reader</b>                          | The course material will consist of lecture videos, handouts of the lectures and computer exercises.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | <p>Exam (50% theory + 50% practice)</p> <p>Passing the course is when the weighted average of the two parts of the exam is equal to or higher than 5.8. The grades of the individual parts of the exam can be lower than 5.8.</p> <p>The exam format will depend on coronavirus restrictions and be communicated in due course (either on site or online).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Special Information</b>             | The teacher can be reached at d.brinks@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Studyload/Week</b>                  | <p>NB2121 is a 3 ECTS course divided over 2 octals for 9 lecture weeks. the course is divided in 9 contact blocks, which are half contact hours with lecturers, and half tutorial with TAs</p> <p>this means an 84 hour time commitment and 9 contact blocks.</p> <p>the students are expected to start each week by watching the lecture videos of that week's topic (2-3 hours). This will be followed by the contact block (2 hours) with the lecturers, which offers the possibility for revision and expansion of the lecture videos, and questions and clarifications. this is followed by the tutorials (2 hours) where the students will work on practical exercises in ImageJ, applying the concepts from the learning videos with help from teaching assistants. this makes for a 7 hour time commitment per lecture week.</p> <p>7x9=63, which leaves 21 hours for exam preparation.</p> <p>The easiest way to go through the course is to keep up with the lecture videos and practical exercises. The contact sessions allow you to ask for clarification about the material found in that week's lecture videos and are most valuable if you have followed the prior lectures. The tutorials allow you to work on the concepts of the week with the help of the TAs, and are most valuable if you have in fact watched the concepts of the week in the lecture videos, and are working on the practical of the week so that the TAs and lecturers can help you in the best way.</p> |   |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Schedule</b> | <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p> <p>Session 1: Introduction - Optical imaging, digitization, quantization, dimensions, pixels, voxels, image displaying, Fourier transform, sampling theory, from analog to digital and back, resolution, noise, signal-to-noise ratio, image file formats.</p> <p>Session 2: Processing - Point operations, histogram equalization, linear filtering, convolution, kernels, separability, smoothing, sharpening, derivatives, edge detection, nonlinear filtering, deconvolution, spatial versus frequency domain.</p> <p>Session 3: Morphology - Binary morphology, structuring elements, erosion, dilation, opening, closing, morphological gradient, reconstruction, ultimate erosion, skeletonization, gray-scale morphology, top-hat filtering, h-dome filtering.</p> <p>Session 4: Segmentation - Intensity thresholding, histogram thresholding, connectivity, connected components labeling, region growing, boundary tracing, watershed transform, minimum-cost tracing, active contours, level sets.</p> <p>Session 5: Quantification - Segmentation quality measures, sensitivity, specificity, ROC analysis, accuracy versus precision, measures of object position, orientation, size, shape, intensities, structure, object classification, colocalization.</p> <p>Session 6: Tracking - Particle detection, particle localization, Gaussian fitting, nearest-neighbor linking, multiple hypothesis tracking, kymograph analysis, cell segmentation, cell tracking, dynamics measures, mean-squared displacement.</p> <p>Session 7: Registration - Alignment problems, marker-based registration, mono/multimodal registration, similarity measures, cross correlation, mutual information, optimization, affine transformation, elastic deformation, image interpolation.</p> <p>Session 8: Visualization - Displaying higher dimensional data, volume rendering, ray casting, maximum intensity projection, alpha blending, surface extraction, mesh generation, lighting, texture mapping, surface rendering, animation.</p> <p>Session 9: Artificial Neural Networks and Deep learning method - Segmentation, classification, logistic regression, peceptrons, training, cost functions, classifiers, Convolutional Neural Networks, U-Net, Deep learning</p> <p>exact content and order of sessions may be updated</p> |
|                 | <b>Location</b> Erasmus MC Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| <b>NB2141</b>                          | <b>Physics 2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>3</b> |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Responsible Instructor</b>          | Prof.dr. C. Joo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Contact Hours / Week</b><br>x/x/x/x | 5/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Exam Period</b>                     | 1<br>2<br>2A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>Course Contents</b>                 | This course provides an introduction to optics, special relativity, quantum mechanics, and other modern physics. Physical principles will be taught with biological examples.                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>Study Goals</b>                     | <p>At the completion of this course the student will be able to:</p> <p>Explain the basic laws of optics, special relativity, and quantum mechanics.</p> <p>Describe physical phenomena with mathematics.</p> <p>Learn how optics and quantum mechanics are used for biophysics research.</p> <p>Apply the basic physical concepts learned here to advanced biology, biophysics and physics topics (Optics, Image Processing and Data Analysis, Statistical Physics, Biophysics, Physical Biology of the Cell, Introduction Nanotechnology).</p> |          |
| <b>Education Method</b>                | <p>Integrated lectures and exercise sessions.</p> <p>The course encompasses integrated lecture/exercise sessions. The lectures cover all chapters of Wolfson about optics, special relativity and quantum mechanics. A detailed overview of the material to be covered in each lecture will be made available at the start of the course.</p>                                                                                                                                                                                                    |          |
| <b>Literature and Study Materials</b>  | <p>Required: Essential University Physics, Richard Wolfson, Addison-Wesley, and Mastering Physics, Richard Wolfson, Addison-Wesley.</p> <p>Supplementary: Principles of Physics, David Halliday, Robert Resnick and Jearl Walker, Wiley, 2011.</p>                                                                                                                                                                                                                                                                                               |          |
| <b>Prerequisites</b>                   | Knowledge of calculus (Analysis 1, 2 and 3), Physics 1, Biophysics                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>Assessment</b>                      | <p>Mid-term exam (50%)</p> <p>Final exam (50%)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>Studyload/Week</b>                  | <p>3 ECTS = 84h</p> <p>~6h/week * 8 weeks = 44h (lectures and exercise classes)</p> <p>It leaves ca. ~4h/week study time at home + 10h exam preparation</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                               |          |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |

| NB2151                                 | Journal Club                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1 |
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| <b>Responsible Instructor</b>          | N. Taneja                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Responsible Instructor</b>          | A. Pines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>Short Course Description:</p> <p>The students will be presented with seminal topics in modern biology (see below). Each topic will be presented during the lectures together with reading assignment consisting of a general review, and some additional material. Moreover, for each topic an experimental paper will be available that will form the basis for the journal club (JC) presentations.</p> <p>The students will be assigned to topic-defined groups and will be required to read/study the corresponding review and additional reading material to acquire a background knowledge on the assigned topic. Next, they will carefully read, discuss the corresponding experimental paper and prepare the JC presentation (exam). Each group will also be asked to lead the Q&amp;A session relative to the JC presentation on a different topic.</p> |   |
| <b>Study Goals</b>                     | <p>The students will learn how to critically read and understand scientific publications, place them in the larger context of a given discipline (the bigger picture!), present their contents, and discuss their merits and flaws. Attention will also be devoted to presentation techniques (use of power point; how to present experimental data; etc.) and to the knowledge of the assigned topic as presented during the lectures and in the reviews.</p>                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | <p>Weekly class working session of 1.5 hours (WC) and of more conventional teaching sessions of 1 hour (HC). Individual and class assignments, sets of questions will be handed in at WCs and HCs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>The scientific articles that form the basis for in class assignments will be posted on Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>NB2151 is a graded course. Evaluation will be based on the journal club presentations and on the Q&amp;A sessions following each presentation. JC assessment will be adjusted by peer-evaluation of each students active participation within each group. Evaluation of the students knowledge of the topics taught at HCs will be integral to the overall assessment.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Special Information</b>             | <p>The teachers can be reached at: a.pines@erasmusmc.nl and n.taneja@erasmusmc.nl</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Location</b>                        | Erasmus MC Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| NB2161                                 | Bioinformatics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4.5 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>          | Dr. J.A. Baaijens                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Instructor</b>                      | A.M.E.T.A. Mahfouz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Instructor</b>                      | Dr.ir. S.J.J. Brouns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Course Contents</b>                 | Bioinformatics is increasingly important to help answer biological questions, to deal with large volumes of measurement data which are now easily generated in the lab in the light of the vast amounts of prior knowledge already present in biological literature and stored in databases. In this course, the student will learn how to use computer methods as tools for biological research. An overview of core application software, databases and online applications will be provided, as well as in-depth knowledge on key algorithms and procedures to process and analyze sequence data and quantitative measurements. The course offers both theoretical insight and practical experience, in computer practicals and projects to be performed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
| <b>Study Goals</b>                     | <p>At the completion of this course, you should be able to:</p> <ol style="list-style-type: none"> <li>1. Describe a range of bioinformatics tasks, tools and databases</li> <li>2. Explain in detail the core bioinformatics algorithms for sequence alignment, statistical testing, clustering and classification</li> <li>3. Argue which algorithms/tools/databases to use for solving a given biological data analysis problem</li> <li>4. Apply bioinformatics algorithms/tools to real biological data</li> <li>5. Interpret the results from a bioinformatics workflow</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>- Lectures</li> <li>- Computer practicals</li> <li>- Projects</li> <li>- Master classes (guest speakers presenting on research in bioinformatics)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Books</b>                           | <p>We will use two books, both of which are publicly available online:</p> <ul style="list-style-type: none"> <li>- Modern Statistics for Modern Biology, by Susan Holmes and Wolfgang Huber (<a href="https://www.huber.embl.de/msmb/">https://www.huber.embl.de/msmb/</a>)</li> <li>- Bioinformatics Algorithms, by Philip Compeau and Pavel Pevzner (<a href="https://www.bioinformaticsalgorithms.org/read-the-book">https://www.bioinformaticsalgorithms.org/read-the-book</a>)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Assessment</b>                      | <p>The course grade is calculated by a weighted average of the Exam grade (60%) and the Project grade (40%).</p> <ul style="list-style-type: none"> <li>- Exam (60% final grade):<br/>One exam (3 hours) covering all the modules of the course, including the Master classes.</li> <li>- Projects (40% of final grade): The projects are graded on a scale 0-10 and require you to hand in a deliverable.</li> <li>- Practicals: Practicals are available per module to practice the material, these assignments are ungraded.</li> </ul> <p>The deadlines for the project deliverables will be presented in the first lecture and are presented in the course schedule in the electronic learning environment.</p> <p>Late policy:<br/>Late submission of a project results in a daily -1 (minus one) on the project grade for every day from the submission deadline until submission.</p> <p>Notes:</p> <ul style="list-style-type: none"> <li>- No scores and/or submissions are transferred to the next year.</li> <li>- In order to pass the course, students must score at least 5.75 on the exam (minimal passing grade).</li> <li>- The resit grade only replaces the exam grade (60% of final grade), not the project grades.</li> <li>- There are no repair opportunities for late submissions.</li> <li>- Exceptions to the late submission policy are possible based on a written request with justification (serious illness, family emergencies) to the responsible teacher, Dr. Jasmijn Baaijens.</li> </ul> |     |
| <b>Studyload/Week</b>                  | <p>NB2161: 4.5 ECTS, 2 octals<br/>4.5 ECTS = 128h<br/>The course has 56 contact hours (lectures and working classes). This leaves ca. 5h/week study time at home + 32h exam preparation.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |

| NB2171                                 | Statistics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. E. Pulvirenti                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Instructor</b>                      | Prof.dr. F.H.J. Redig                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/8/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 3<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 3<br>3A<br>3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>This course covers the basics of probability theory and statistics.</p> <p>Topics in probability include discrete and continuous random variables, probability distributions, and stochastic independence, joint distributions, sums and functions of random variables, expectation, variance and covariance, law of large numbers, and central limit theorem.</p> <p>Topics in statistics include random sample and sample distributions, estimators (bias, error, and maximum likelihood), least squares and confidence intervals for the average of normally distributed data with known or unknown variance, and hypothesis tests.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Study Goals</b>                     | <ol style="list-style-type: none"> <li>1. The student knows the most important discrete and continuous distributions and some contexts in which they arise. For distributions in terms of a density or mass function, the student can compute expectations and variances.</li> <li>2. The student can apply the Theorem of Bayes to compute conditional probabilities. The student can apply the notion of independence.</li> <li>3. The student can apply Markov's, Chebyshev's and Jensen's inequality, and knows in which context these inequalities are of use.</li> <li>4. The student can apply the law of large numbers and the central limit theorem.</li> <li>5. The student can compute whether an estimator is biased, and compute its efficiency. The student can compare different estimators.</li> <li>6. The student can compute the maximum likelihood estimator and can apply the invariance principle.</li> <li>7. The student can perform a hypothesis test and can construct confidence intervals in the context of concrete data-sets. The student can compute errors of the first and second kind.</li> </ol> |   |
| <b>Education Method</b>                | Lectures and exercise classes [exact format may depend on the current Corona regulations at the time of the course]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Books</b>                           | <p>A Modern Introduction to Probability and Statistics Understanding Why and How<br/> Springer Texts in Statistics, Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. 2005, XVI, 488 p. 120 illus.,<br/> Hardcover, ISBN: 1-85233-896-2</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                      | <p>On campus, 3 hours exam.</p> <p>Disclaimer: information may change depending on the developments around the coronavirus.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Studyload/Week</b>                  | <p>3 ECTS, 1 octal, 3 ECTS = 84h<br/> 13 lectures * 2h/lecture = 26 h (lectures)<br/> study pre-lecture and pre-lecture exercises 1h -&gt; (13h in total)<br/> Leaves 25h study time and exercises from lectures and 20h exam preparation<br/> Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Location</b>                        | TU Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |



| NB2181                                 | Computational Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Ir. M.J.F. Negrello                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/4/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 3<br>4<br>3A<br>3B<br>4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Required for</b>                    | This course delivers essential knowledge for all scientific computing courses in the nanobiology program and in a professional career.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | The student learns to use modelling and simulation techniques in science and engineering, with special emphasis on nanobiology. The students first become familiar and/or improve their existing skills with the programming environment of Python. The students work then on assignments (individually or in groups) where they learn different numerical methods and their implementation in Python. At the end of the course the students use computer modelling in order to analyse and solve a scientific problem related to nanobiology.                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>At the end of this course</p> <ol style="list-style-type: none"> <li>1. The student can analyze, manipulate and visualize data in the python programming language, and specifically, in the jupyter notebook environment.</li> <li>2. The student can translate mathematical statements and physical models into numerical algorithms into working computer code.</li> <li>3. The student can reason about the effect of data resolution, numerical precision, convergence, computer memory and computational speed on the numerical solution of a scientific or engineering problem.</li> <li>4. The student can combine computational tools in linear algebra, random numbers, calculus into a computational model of physical phenomena.</li> </ol> |   |
| <b>Education Method</b>                | <p>- Practical activity where students solve python projects on basic skills necessary for scientific computing.</p> <p>Skills acquired:</p> <ul style="list-style-type: none"> <li>- Translating mathematical models into computational algorithms</li> <li>- Debugging</li> <li>- Explaining own code</li> <li>- Commenting code</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Assessment</b>                      | Normal assessment: An electronic exam worth 60% of the mark and a final project with oral assessment worth 40% (both have separate resits). In the case of off-campus exams, the assessments may be modified.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Special Information</b>             | The teacher can be reached at <a href="mailto:m.negrello@erasmusmc.nl">m.negrello@erasmusmc.nl</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Studyload/Week</b>                  | <p>In addition to lecture hours, students are expected to invest an extra 4h/week honing their coding skills.</p> <p>Students with little to no previous coding knowledge should expect to invest up to 8h/week.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                               |   |
| <b>Location</b>                        | TU Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| NB2191                                 | Differential equations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. B. van den Dries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 5/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                     | 1<br>2<br>1A<br>1B<br>2A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | Differential equations occur in all branches of science and engineering as mathematical models for reality. Answering questions that may pop up in any of these branches often comes down to analysing and solving such differential equations.<br>The aim of this course is to introduce you to a variety of techniques to solve differential equations of several common types. In addition, we will discuss techniques to extract information from differential equations without solving them.                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Study Goals</b>                     | <ol style="list-style-type: none"> <li>1. For a given (system of) differential equations the student can tell whether it is linear / non-linear, homogeneous / non-homogeneous, separable, exact or partial.</li> <li>2. The student can make statements about existence, uniqueness and analyticity of solutions for boundary value problems and initial value problems without explicitly solving.</li> <li>3. the student can make statements about qualitative properties of solutions such as limiting behavior, behavior near equilibrium points and radius of convergence of power series, without explicitly solving.</li> <li>4. The student can apply various methods for constructing solutions to (a system of) DEs: Separation, variation of constants, reduction of order, using potentials, Laplace transform, series solutions, Fourier series.</li> </ol> |   |
| <b>Education Method</b>                | Lectures (4 hours per week) and instruction classes (2 hours per week). In principle all activities will be on campus.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Books</b>                           | William E. Boyce & Richard C DiPrima, Elementary Differential Equations and Boundary Value Problems, Eleventh Edition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | One midterm (40%) and endterm(60%), no bonus regulation. Details will be provided at the beginning of the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Studyload/Week</b>                  | <p>There are no separate resits for midterm and endterm, only one resit for the full course with weight 100%.</p> <p>This is a 3EC course, which comes down to 84 hours.</p> <p>Contact hours: 48 hours (16 2-hour lectures + 8 2-hour exercise classes)<br/>Homework (outside exercise classes): 20 hours (2.5 hours per week, 8 weeks)<br/>Exams: 16 hours (5 hours exams, 11 hours preparation)</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                              |   |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |

| NB2214                                 | Electronic Instrumentation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. M.W. Docter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 6/6/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Required for</b>                    | the Python programming will be continued in Bioinformatics NB2161 and Computational Science NB2181.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Expected prior knowledge</b>        | <p>the Fourier transforms and Aliasing will be more quantitatively explained in Signals and Systems TN2545.</p> <p>- Pre-knowledge from the first year Nanobiology are the Physics 1b NB1143 (electromagnetism), BioPhysics NB1132 (Python programming), and Differential Equations NB2191</p> <p>- It is recommended to follow Signals and Systems TN2545 at the same time as the Electronics Instrumentation course. Some concepts as Fourier Domain and Aliasing are treated qualitatively in EI, and quantitatively in S&amp;S</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>Octal 1: Passive Electronics:<br/>RLC, network theory (including voltage dividers, Norton/Thevenin circuits, superposition), filtering Equipment (multimeter, function generator, oscilloscope, scope probe, triggering, quantization), Analysis (uncertainty, Fourier)</p> <p>Octal 2: Programming &amp; Data Acquisition<br/>Refreshing Python programming(calculations, for loop, if statement, comparison) by challenging exercises<br/>Basics Data Acquisition (Analog &amp; Digital I/O)<br/>Measurement automation (turn a motor, voltage response of a filter)<br/>(anti) Aliasing, digital filtering</p> <p>Octal 3: Active Electronics<br/>Diodes<br/>Noise &amp; interference<br/>Simple opamp circuits: integrator, differentiator<br/>Bridge circuit, instrumentation amplifier</p> <p>Octal 4: Active Electronics (Sensors &amp; Transistors) &amp; Final Assignment (both two weeks)<br/>Sensors like photodiode, capacitive &amp; temperature sensor<br/>Comparator, oscillator<br/>FET and transistor<br/>Final Assignment: apply all knowledge; configure your own colorimeter and measure a biological process.</p>                                                                                                                                                                                                                                                                                       |   |
| <b>Study Goals</b>                     | <p>After successful completion of this course students are capable of:</p> <p>Designing and conducting a basic electronic measurement by hand or via computer<br/>..o Have elementary skills to use electronic measuring equipment, data acquisition software, control instruments using a computer interface.<br/>..o Be able to communicate on a professional level with electronics experts and design a good measuring system with these experts.<br/>..o Be able to use a computer to control a measuring system, read data and analyse these data</p> <p>Understand the influences of the electronics on their measurement outcome<br/>..o Understand the working mechanism of basic components (passive and active, and their use in filters, sensors, amplifiers, etc)<br/>..o Know how to optimize the quality of the measured signal<br/>..o Understand the most important causes of noise and interference and is able to analyse the influence of this on a measured signal</p>                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                | <p>the education method is blended learning. This means learning is not limited to the contact hours:<br/>2 hours instruction (theory+exercises) per week,<br/>4 hours Practical sessions using electronics set-up per week</p> <p>You are expected to prepare the theory before the instruction session, and make homework (i.e. predictions of the practical assignments) before the practical session</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Computer Use</b>                    | <p>- For data acquisition in the SCR we use the fixed desktop computer there, which contains a Data Acquisition card.<br/>- All lab assignments work with online Jupyter Notebooks (on a Vocareum website, accessible through BrightSpace).<br/>- For accessing and working with these digital notebooks, students use their own laptop.<br/>- For use of the ALPACA (electronics board), students run a local copy of the notebook, and connect the ALPACA with an USB 2 connector (Mac users bring an adapter)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Literature and Study Materials</b>  | <p>The practicum manual is given in the form of online Jupyter Notebooks (Vocareum, via BrightSpace).</p> <p>The slides, videos and other additional materials are provided through BrightSpace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | <p>Book: Electronics: A Systems Approach (6th edition), author: Neil Storey, Pearson Education Limited</p> <p>The final grade is an average over 4 intermediate grades: 3 exams and a report, each counts for 25%.<br/>- Three intermediate exams: 1 hour exams in week 1.5, 1.10, 2.5.<br/>- Resits to be announced (most likely in 1.10, 2.5, 2.10)<br/>- The report is about the fourth octal, with focus on the final assignment in week 2.8 and 2.9, and the deadline will be end of week 2.10.<br/>- You can only join an exam after finishing all corresponding practicals (checked off by a TA).</p> <p>You pass the course when you have done all exams and reports, and your overall score is 5.8 or higher.<br/>- Individual exam grades might be below 5.8 and compensated with other exam grades.<br/>- The highest score on each exam counts<br/>- Grades are granted after returning the Alpaca (Arduino (based) Learning Platform for Analog Circuits and Automation, which you will get on loan during the course).</p> <p>- Students, who have missed or not completed any assignment, must attend an additional catch-up session which will be scheduled once per octal, previous to each exam.<br/>- all practical work requires individual preparation, and lab work in groups of 2 students<br/>- Exams are done individually, the report as well (but the final assignment is performed in a group).</p> |   |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Permitted Materials during Tests</b> | <p>the intermediate exams are digital, meaning they are on a TUDelft computer on campus. Only there a shielded environment (no internet) with access to only our files is possible.</p> <p>You are provided with:</p> <ul style="list-style-type: none"> <li>- computer (including questions, formula sheet and calculator)</li> <li>- scrap paper</li> </ul> <p>You can bring with you:</p> <ul style="list-style-type: none"> <li>- Calculator (simple one)</li> <li>- pencil</li> </ul> <p>It is a close book exam.</p>                                                                                                                                                                                                                                      |
| <b>Studyload/Week</b>                   | <p>6ECTS = 168h</p> <p>16 weeks * 2 hours (lectures and instruction)= 32 h</p> <p>16 weeks * (4 hours practicum + 0.5 hours preparation) = 72 h</p> <p>3 exams * (1.5 hours exam + 16 hours preparation)=52.5 h</p> <p>1 report * 12 hours =12 hours</p> <p>total: 168.5</p> <p>not yet included optional hours:</p> <p>4 exam weeks * 4 hours catch-up practicum (only if you have not yet finished the practicum during regular hours)</p> <p>4 exam weeks * 2 hours exam practice</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p> |
| <b>Location</b>                         | <p>Instruction sessions of 2 hours (location to be announced, a regular instruction classroom in Delft).</p> <p>Practical sessions of 4 hours in Studio Classroom 1/2/3, AS building 22 (Lorentzweg1, A151) Delft</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| NB2220                                 | Statistical Physics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. J.W. Zwanikken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | <p>This course will give an introduction to statistical physics, covering the basic concepts of: emergent phenomena, the random walk, microstates and macrostates, the ideal gas, the ideal polymer, two definitions of entropy, temperature, thermodynamic potentials and ensembles, the Boltzmann distribution, the partition function, phase transitions, electrostatic screening, and the Ising model. Knowledge will be applied to predict the elastic properties of ideal polymers, the barometric height distribution, the distribution of molecules in a potential landscape (in particular, an optical tweezer), the critical temperature of a fluid, and the coexisting densities of a bi-stable phase (with a focus on liquid-liquid phase separation).</p> <p>Expected prior knowledge:<br/>NB2011 Thermodynamics and Transport</p> |   |
| <b>Study Goals</b>                     | <p>Learning objectives (LO)</p> <ol style="list-style-type: none"> <li>1. Understanding the topics listed under 'Course Contents', and how they are conceptually connected</li> <li>2. Being able to analytically solve both elementary and advanced problems using the standard procedures of the theory.</li> <li>3. Apply the concepts to get a deeper conceptual understanding of (living) materials.</li> <li>4. Write clear and detailed solutions to problems, such that they can be understood by your peers.</li> </ol>                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | <p>Lectures: LO1, LO2, LO3</p> <p>Homework: LO1, LO2, LO3</p> <p>Exercise classes: LO1, LO2, LO3</p> <p>Peer grading: LO4</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | <p>To pass you always need to score at least 5.8 on the final exam.</p> <p>The final grade will be based on the final exam, plus other elements if they have a favorable effect on the final grade. Their relative weight is expressed in terms of points:</p> <p>Homework: The best 7 out of 8 HW sets will be counted if they increase the overall grade, with a maximum of 50 points</p> <p>Attendance: Will be counted if it increases the overall grade, with a maximum of 20 points</p> <p>Group assignments (2): Will be counted if they increase the overall grade, with a maximum of 30 points</p> <p>Midterm exam: Will be counted if it increases the overall grade, with a maximum of 100 points</p> <p>Final Exam: 300 points</p> <p>@Retakers<br/>old homework will not count towards the final grade.</p>                        |   |
| <b>Studyload/Week</b>                  | <p>There will be 4 hours of lecture and 4 hours of work session per week, which should cover the entire study load per week. Preparation for the exams will take extra time, roughly 8 hours each + error bar.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Location</b>                        | TU Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |

| NB2230                                 | Biomolecular Structure and Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. J.H.G. Lebbink                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 1<br>2A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Expected prior knowledge</b>        | NB1102 Chemistry 1; NB1012 Biochemistry; NB1016 Molecular biology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | This course teaches the basic determinants and elements of biomolecular structure and dynamics, structure-function relations, and current methods in structure determination, with a focus on proteins and their complexes with other biomolecules. It furthermore provides practical information on how to navigate the publicly available structural databases and how to interpret, analyze and optimally visualize biomolecular structures.                                                                                                                                                                                |   |
| <b>Study Goals</b>                     | <p>At the completion of this course the student will be able to:</p> <ul style="list-style-type: none"> <li>- Describe energetic principles, composition and elements that define biomolecular structure</li> <li>- Describe relations between structure, dynamics, interactions and function</li> <li>- Have insight into current methods for macromolecular structure determination</li> <li>- Be able to navigate structural databases</li> <li>- Be able to critically analyze biomolecular structures</li> <li>- Be able to create high-quality figures containing detailed biomolecular structure information</li> </ul> |   |
| <b>Education Method</b>                | The course will be taught using lectures and guest lectures, computer assignments and homework to be handed in and discussed in class. Professional skills that will also be developed are reading of scientific papers, integrating and applying scientific knowledge, problem solving and time management.                                                                                                                                                                                                                                                                                                                   |   |
| <b>Computer Use</b>                    | yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Literature and Study Materials</b>  | Recommended Book: The Molecules of Life by John Kuriyan ISBN: 9780815341888                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | Class and Homework Assignments (20%), midterm exam (25%), final exam (55%). Format may change due to COVID-19 restrictions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Enrolment / Application</b>         | For course: Brightspace. For exam: Osiris                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Special Information</b>             | The teacher can be reached at j.lebbink@erasmusmc.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Studyload/Week</b>                  | <p>3ECTS = 84 hours = 8.4 hours/week<br/> Lectures and practicals 4 hrs/week * 8 weeks = 32 hours<br/> Homework 2 hrs/week * 8 weeks = 16 hours<br/> Exams (midterm and final) 3 hrs/week * 2 weeks = 6 hours<br/> Study course material at home 3hrs/week * 10 weeks = 30 hours</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                    |   |
| <b>Location</b>                        | Erasmus MC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |

| TN2545                                 | Systems and Signals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 6 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. D.J. Verschuur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr. F.M. Vos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/12/12/0/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                | 2A<br>2B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                 | 2A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 2A<br>2B<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Expected prior knowledge</b>        | Complex numbers, elementary calculus, including derivatives & integrals, solving linear equations with few unknowns, geometric series, trigonometric & exponential functions. Create and analyze graphs of elementary functions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | Systems and signals in continuous and discrete time. Basic deterministic signals. Convolution and properties of LTI systems. Fourier series and Fourier transforms with their properties. Filters in theory and practice: first- and second-order DE, Butterworth filters, Gaussian derivative filters. The estimation of spectra in a time window and the associated uncertainty relation. The DFT and FFT as practical tools. Introduction to 2D Fourier transforms. Modulation and demodulation techniques, the analytic signal and Hilbert transforms. Sampling (Nyquist theorem), aliasing and reconstruction. Laplace and Z-transforms with their ROC and properties. Feedback systems and associated stability determination.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Study Goals</b>                     | <p>The main goal for this course Systems and Signals is to learn to analyze linear, time-invariant systems and associated signals and apply mathematical operations and techniques on these systems and signals at a 2nd year BSc level relevant to, i.a. electronics, optics, acoustics, mechanics and physical processes.</p> <p>To attain this main goal, the following learning goals have been formulated.</p> <p>After successfully engaging in S&amp;S (TN2545), a student should be able to:</p> <ol style="list-style-type: none"> <li>1. recognize and determine basic properties of systems including linearity, time invariance, impulse response, and determine its response based on convolution.</li> <li>2. derive Fourier series coefficients and Fourier transform of elementary signals.</li> <li>3. describe the effect of frequency-dependent filtering of discrete and continuous-time signals and relate this to linear difference/differential equations with constant coefficients.</li> <li>4. analyze the effect of sampling of band-limited signals and reconstruct a continuous-time signal from the discrete samples.</li> <li>5. perform amplitude-modulation to elementary signals and derive and analyze associated reconstruction methods.</li> <li>6. assess properties of systems based on the Laplace and Z-transform.</li> </ol> <p>All of the above techniques are taught and assessed analytically and, when applicable, graphically.</p> |   |
| <b>Education Method</b>                | Oral lectures by teachers, practical working classes with supervision                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | A.V. Oppenheim and A.S.Willsky, with S.H.Nawab, Signals and Systems, Prentice-Hall, 2nd edition, Ch. 1 to 11 (obtainable in bookshop and VVTP).<br>Reader with additional material en lecture notes/hand-outs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | Combination of (bonus) tests and final exam.<br>See Brightspace for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |

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|---------------------|-----------------------------|
| <b>Year</b>         | <b>2022/2023</b>            |
| <b>Organization</b> | <b>Applied Sciences</b>     |
| <b>Education</b>    | <b>Bachelor Nanobiology</b> |

### 3e jaar BSc NB 2022

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|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsible Program Employee</b> | J.C. Colgrove                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Contact for Students</b>         | Students with questions about minors or electives should contact the Academic Counselor <Studieadviseur-NB@tudelft.nl>                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Introduction 1</b>               | <p>The 3rd year of the programme consists of 3 key parts.</p> <ul style="list-style-type: none"> <li>* A one semester long minor, 30 EC</li> <li>* The Bachelor End Project (Thesis), 20 EC</li> <li>* Nanobiology Electives, 10 EC</li> </ul> <p>The minor is scheduled in the first semester of the third year. Nanobiology students may choose any available minor offered by TU Delft, Erasmus MC, or Leiden University after approval from the Board of Examiners a minor at a different university or a self-composed minor (including minors abroad).</p> |

| NB3000                                 | Bachelor Thesis Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 20 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | D. Mevius                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Responsible Instructor</b>          | J.C. Colgrove                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Course Coordinator</b>              | L. van der Elst                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| <b>Course Coordinator</b>              | Drs. E. Jacobs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | self study, approximately 28 hours per week.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |
| <b>Education Period</b>                | 3A<br>3B<br>4A<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| <b>Start Education</b>                 | 3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| <b>Exam Period</b>                     | Exam by appointment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |
| <b>Course Contents</b>                 | The Bachelor End Project (BEP) is the last part of your Bachelor of Science in Nanobiology. In this project the knowledge and skills learned in the first two years is applied on a level that enables you to successfully follow a connecting Master programme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| <b>Study Goals</b>                     | <p>Student is able to</p> <ol style="list-style-type: none"> <li>1. Understand and directly apply theoretical knowledge to research</li> <li>2. use scientific methods and approaches in the areas of responsibility, communication, literature study, critical attitude and time planning.</li> <li>3. Do research work including developing significant original ideas, and improving experimental setups including for safety.</li> <li>4. Write an accurate independent report of a quality that it can be used for further research.</li> <li>5. Present a clear, well-argued defense of their project and able to answer questions about it.</li> <li>6. Be an effective team member with good English writing skills. Student shows open mindedness and creativity.</li> </ol> <p>The full grading scheme can be found at the TNW Eindprojecten Brightspace page.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |
| <b>Education Method</b>                | Individual project in a research group at TUDelft or ErasmusMC. Activities include independent research, lab meetings, writing, presenting, data analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
| <b>Assessment</b>                      | <p>- A report (see document BEP thesis guidelines in Brightspace page for Thesisoffice TNW)</p> <p>- A final presentation usually 20-30 min open to the public.</p> <p>- A defense. A 30-45 minute closed session with student and assessment committee where student can defend their thesis.</p> <p>This committee will consist of at least two examiners that are member of scientific staff (tenured or tenure track).</p> <ol style="list-style-type: none"> <li>1. The responsible supervisor is one of the examiners.</li> <li>2. Both examiners must be on the Nanobiology GreenList, one must be a teacher in the Nanobiology bachelor or master programme as indicated on the GreenList.</li> </ol> <p>The teaching member of the assessment committee must be identified at the time of application submission.</p> <p>The GreenList is available from the TNW EindProjecten BrightSpace page.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Enrolment / Application</b>         | To start this project the student must have completed all year 1 courses and at least 60 EC from year 2 and 3. The BEP enrollment form must be submitted and accepted by the Thesis office before beginning the project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Remarks</b>                         | <p>Check on the Brightspace page of the endproject administration / Thesis office TNW for detailed information and required forms.</p> <ol style="list-style-type: none"> <li>1. A Bachelor End Project earns the student 20 ECTS and is usually completed in one semester consisting of 20 weeks.</li> <li>2. This is planned for semester 2 of year 3 (from Second week in February to First week in July usually), but can be started at a different time depending on study progress.</li> <li>3. To start this project the student must have completed all yr 1 courses and at least 60 points from year 2 and 3.</li> <li>4. Students are responsible for finding an available supervisor. The supervisor must be on the Nanobiology "GreenList" which is available on the Thesis Office BrightSpace page. Together the student and supervisor decide on a project. The student then completes and submits the BEP Approval form. This is checked by the programme coordinator or study advisor before the student is allowed to start.</li> <li>5. The projects should be research work within an existing project in a lab. They should have substantial practical and theoretical components and lay the foundation for eventual independence.</li> <li>6. The project is completed by a written report (BSc) thesis and oral evaluation procedure.</li> <li>7. The student is expected to take part in research work about 28 hours per week. This includes participating in lab meetings, journal clubs and the like, as scheduling permits, while also completing their elective courses.</li> </ol> <p>Note that the end grade will only be registered when your (digital) end-report is handed in to the Thesis office. You will automatically receive a digital survey.</p> <p>Info via: ThesisOffice-TNW@tudelft.nl</p> <p>Thesis Office is located at the TN building, room A208</p> |    |



|                     |                             |
|---------------------|-----------------------------|
| <b>Year</b>         | <b>2022/2023</b>            |
| <b>Organization</b> | <b>Applied Sciences</b>     |
| <b>Education</b>    | <b>Bachelor Nanobiology</b> |

### 3e jaar BSc NB keuzevakken 2022

|                                     |               |
|-------------------------------------|---------------|
| <b>Responsible Program Employee</b> | J.C. Colgrove |
|-------------------------------------|---------------|

|                       |                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction 1</b> | Students must take at least 10 EC of electives. Please see the study guides for each course for information about enrollment requirements. Some electives have limited enrollment, and are generally offered multiple times to provide room for all students. Instructions on how to enroll are included on the study guide for those electives. |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| NB3011                                 | Nanomedicine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2.5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>          | Dr. T.L.M. ten Hagen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/4*/4*/0/4*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| <b>Education Period</b>                | 3<br>4<br>3A<br>3B<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Start Education</b>                 | 3<br>4<br>3A<br>3B<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Exam Period</b>                     | 3<br>4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Course Contents</b>                 | The course is particularly focused on the application of nanocarriers for cancer therapy. Emphasis will be on combination of these carriers with external triggers to improve applicability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Study Goals</b>                     | <p>Students will</p> <ol style="list-style-type: none"> <li>1) learn the composition, construction and manipulation of lipid-based nanocarriers. Next to the production of standard carriers targeted and trigger-sensitive nanocarriers are discussed.</li> <li>2) learn about the application of lipid-based nanocarriers in human disease.</li> <li>3) learn about tumor pathophysiology and factors influencing nanocarrier functionality.</li> </ol> <p>At the completion of this course the student will be able to:</p> <ol style="list-style-type: none"> <li>1. Position nanocarriers with respect to other particulate matters</li> <li>2. Reproduce essential building blocks of lipid-based nanocarriers</li> <li>3. Understand possible ways to manipulate nanocarrier functionality</li> <li>4. To build a standard lipid-based nanocarrier</li> <li>5. Provide information on the use of nan-carriers in cancer therapy</li> <li>6. Interpret aspects of tumor pathophysiology which influence nanocarrier performance</li> <li>7. Understand limitations and shortcomings of drug delivery to solid tumors</li> <li>8. Provide possible methods using nanocarriers to improve drug delivery to solid tumors</li> </ol> |     |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>- Lectures.</li> <li>- Practical</li> <li>- subgroups</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Literature and Study Materials</b>  | -Selected publications; mainly articles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Books</b>                           | -Selected publications; mainly articles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Assessment</b>                      | <p>Pass / fail. Nanomedicine elective is an elective; you can chose to enroll or not.</p> <p>Program is such that days follow each other with an end result on the last day. Days are filled with presentations and hands on lab work. Missing one day effects the next. Last day is results presentation, report presentation and presentation of assignment. Attending is not mandatory, but to receive a grade all days need to be followed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| <b>Enrolment / Application</b>         | This course is 5 weeks long and is offered three times--in periods 3a, 3b and 4b. Space is limited and is offered on a first come, first served basis with priority given to Nanobiology students. Registration for a place begins after November 1 and continues until all spaces are alloted and takes place in the Nanobiology Bachelor Brightspace page. If you                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| <b>Special Information</b>             | The teacher can be reached at t.l.m.tenhagen@erasmusmc.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| <b>Studyload/Week</b>                  | Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Location</b>                        | Erasmus MC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |

| NB3014                                 | Computational Neuroscience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2.5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>          | Ir. M.J.F. Negrello                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| <b>Education Period</b>                | 4<br>4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| <b>Start Education</b>                 | 4<br>4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| <b>Exam Period</b>                     | 4<br>4A<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| <b>Course Contents</b>                 | Biologically inspired neural networks perform a range of computational functions analogous to cognitive processing in the living brain, e.g., as pattern recognition, associative learning, motor control, forecast and prediction. In this course we will describe basic principles and algorithms underlying these functions. With a sandbox neural-network construction software we will implement these networks and familiarize with their properties and functions.                                                                                                  |     |
| <b>Study Goals</b>                     | <p>At the completion of this course the student will be able to:</p> <ol style="list-style-type: none"> <li>1. Produce biophysically accurate neuronal models.</li> <li>2. Interpret neuronal activity with dynamical systems theory.</li> <li>3. Produce spiking networks.</li> <li>4. Explain how neurons may recognize input patterns through the perceptron analogy</li> <li>5. Train a neural network to perform pattern recognition.</li> <li>6. Intuitively understand how recurrent connections of a recurrent neural network leads to complex dynamics</li> </ol> |     |
| <b>Education Method</b>                | <p>Computational projects (python) and Lectures</p> <p>Individual/group programming assignments</p> <p>Skills acquired in this course:</p> <ul style="list-style-type: none"> <li>- Reading and writing python code (from the python projects)</li> <li>- Obtaining and presenting new knowledge (acquired in the final project)</li> <li>- Giving reviews (on peer's projects)</li> <li>- Writing report</li> <li>- Presenting</li> <li>- Team work</li> </ul>                                                                                                            |     |
| <b>Books</b>                           | <p>Educational material provided as jupyter notebooks.</p> <p>Selected articles will be provided.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| <b>Prerequisites</b>                   | <ul style="list-style-type: none"> <li>- biology of the cell</li> <li>- biophysics of cell membranes (membrane potential, concentration)</li> <li>- python (numpy, matplotlib)</li> <li>- linear algebra (dot and cross products, eigenvectors and eigenvalues)</li> <li>- calculus (differential equations)</li> </ul>                                                                                                                                                                                                                                                    |     |
| <b>Assessment</b>                      | <p>Students will be peer-evaluated via a project presentation (50%) and a report (50%)</p> <p>Projects will be groups of 1 or 2. Reports and presentations are given by the group.</p> <p>Students get one joint grade for the presentation and an individual grade for your report.</p> <p>Students pass if the combined grade meets the minimum.</p> <p>RESIT:</p> <ul style="list-style-type: none"> <li>- Deliver own code implementing a neuronal model and a 3000 word report about its limitations/improvements/assumptions.</li> </ul>                             |     |
| <b>Special Information</b>             | The teacher can be reached at <a href="mailto:m.negrello@erasmusmc.nl">m.negrello@erasmusmc.nl</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Tags</b>                            | <p>Abstract</p> <p>Artificial intelligence</p> <p>Lineair Algebra</p> <p>Mathematics</p> <p>Modelling</p> <p>Numeric Methods</p>                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Studyload/Week</b>                  | <p>4h/week * 5 weeks = 20 h (lectures and working classes)</p> <p>ca. 8h preparation of final presentation and report.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                          |     |
| <b>Location</b>                        | Erasmus MC Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |

| NB3015                                 | A primer in Neuroscience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2.5 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>          | Ir. M.J.F. Negrello                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Education Period</b>                | 4<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| <b>Start Education</b>                 | 4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
| <b>Exam Period</b>                     | 4<br>5<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| <b>Course Contents</b>                 | <p>NB3015 is a flip-the-classroom experience, with emphasis on developing the ability to read and grasp essentials from neuroscience topics, while emphasizing a mechanistic understanding of brain function. We will be reading chapters from "Principles of Neuroscience" (5th Edition) and students will present them in a pre-established sequence</p> <p>Week 1. Introduction and Plan of Attack (Lecture)</p> <p>Week 2. Sensation and Perception (Student Presentations)</p> <p>Week 3. Sensorimotor Behavior (Student Presentations)</p> <p>Week 4. Cognition (Student Presentations)</p> <p>Presentation Rules</p> <p>- Presentations will last for 20'+5-10' (Q&amp;A) and will involve groups of maximum 2 people. Note: Depending on the number of students taking the course the presentation time allotted may vary.</p> <p>PS. Though not mandatory, NB3014 (Computational Neuroscience) is a helpful pre-requisite.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| <b>Study Goals</b>                     | <p>Learning Goals</p> <p># Neurons and Synapses</p> <ul style="list-style-type: none"> <li>- [ ] Know different kinds of cortical neurons and their distinguishing features.</li> <li>- [ ] Know the principal (structural and functional) differences between excitatory and inhibitory synapses.</li> <li>- [ ] Know the principal neurotransmitters of the brain.</li> <li>- [ ] Give examples of processing hierarchies in the nervous system.</li> <li>- [ ] Describe connectivity patterns between neurons in the cortex.</li> </ul> <p># Sensation and Perception</p> <ul style="list-style-type: none"> <li>- [ ] State the goal of the study of psychophysics.</li> <li>- [ ] Be able to explain what are psychophysical laws.</li> <li>- [ ] Describe probabilistic strategies to quantify perception.</li> <li>- [ ] Give examples of sensory features encoded by neurons.</li> <li>- [ ] Know the role of the somatosensory systems in the control of behavior.</li> </ul> <p># Perception and Movement</p> <ul style="list-style-type: none"> <li>- [ ] Describe the central problem of cognitive science.</li> <li>- [ ] Explain what is a neural map and how it is obtained.</li> <li>- [ ] Know basic differences between the sensory and motor somatotopic maps.</li> <li>- [ ] Be able to describe what is an internal model.</li> <li>- [ ] Be able to differentiate between forward and inverse models.</li> <li>- [ ] Be able to explain what is segmentation.</li> <li>- [ ] Understand the hierarchical organization of visual processing.</li> </ul> <p># The Organization of Cognition</p> <ul style="list-style-type: none"> <li>- [ ] Be able to produce a definition of cognition.</li> <li>- [ ] Be able to give examples of serial and parallel processes in the brain.</li> <li>- [ ] Be able to explain why receptive fields size grows during information processing by the brain.</li> <li>- [ ] Be able to explain what are association cortices, and name examples.</li> <li>- [ ] Understand the inextricable relationship between action and perception in the sensorimotor loop.</li> <li>- [ ] Be able to describe the different kinds of memory.</li> <li>- [ ] Be able to name the molecular mechanism responsible for long term memory storage.</li> <li>- [ ] Know the differences between imaging methods in terms of goals</li> <li>- [ ] Explain how neural correlates of consciousness be measured with imaging.</li> </ul> |     |
| <b>Education Method</b>                | <p>Students read and summarize selected chapters of "Principles of Neural Science" according to specific learning goals.</p> <p>Skills:</p> <ul style="list-style-type: none"> <li>- Reviewing, summarizing and presenting complex knowledge</li> <li>- Receiving and giving peer feedback</li> <li>- Writing a short scientific report</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
| <b>Books</b>                           | <p>Kandel - Principles of Neuroscience<br/> Gerald Edelman - Topobiology<br/> Ross Ashby - Design for a Brain<br/> Varela and Maturana - The tree of knowledge<br/> Llinas - I of the Vortex</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Prerequisites</b>                   | NB3014 (recommended)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Assessment</b>                      | <p>Evaluation Rules (NB3015):<br/> The evaluation is based on 2 components (50/50). Students pass if they achieve the combined passing grade.</p> <p>1) The presentation grade:</p> <p>1.1 The grade of a student presentation where 30% of the grade is from 'peer-evaluation', and 70% is due to lecturer evaluation.</p> <p>1.2 The evaluation of presentations (peer and lecturer) is done according to four dimensions: Form (slides), Content (selection of</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>material), Clarity (of explanation), Q&amp;A.</p> <p>1.3 Grade for presentation is shared equally by the two presenters</p> <p>2) An essay:</p> <p>2.1 The essay focuses on a different topic than your presentation. From all of the other presentations, you select a topic that has piqued your interest and expound on it. The essay can take the form of a critique, a different question, an observation, a proposed hypothesis, an experimental question.</p> <p>2.2 Suggested word count: 2000.</p> <p>2.3 Grade will depend on correctness, clarity and concision.</p> <p>RESIT: A 5000 word essay combining two of the possible presentation topics.</p> <p>The teacher can be reached at <a href="mailto:m.negrello@erasmusmc.nl">m.negrello@erasmusmc.nl</a></p> |
| <b>Special Information</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Studyload/Week</b>      | <p>In class we have 4h per encounter, for a total of 5 encounters, thus 20h. Extra class hours amount for about 4h for the group that is preparing presentations. Reviewing the week's material should take about 4h per week. Writing the final report should take about 4h.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                        |
| <b>Location</b>            | Erasmus MC Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| NB3016                                  | High Speed Scientific Computing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2.5 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>           | Dr.ir. C. Strydis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| <b>Responsible for assignments</b>      | Dr.ir. C. Strydis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/0/12/0/0/0/                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| <b>Education Period</b>                 | 3<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Start Education</b>                  | 3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| <b>Exam Period</b>                      | Exam by appointment<br>3A<br>3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Expected prior knowledge</b>         | Python programming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| <b>Course Contents</b>                  | Scientific computing is a new field of science making use of computers to analyze and solve scientific problems. It involves construction of mathematical models which - for the first time - permit quantitative analysis of complex phenomena. While an excellent tool for analysis, scientific computing alas comes at a price: Complex simulations generally mean slow simulations. Drawing illustratory material from various scientific domains, this course will teach students hands-on the basics of algorithmic complexity, the tricks behind constructing efficient (parallelizable) algorithms and will introduce them to machines capable of executing high-speed simulations, which are orders of magnitude faster than existing approaches.                                                                                                                                                                                                                                                                                            |     |
| <b>Study Goals</b>                      | <p>At the completion of this course the student will be able to:</p> <ol style="list-style-type: none"> <li>1. Understand the basics of scientific computing.</li> <li>2. Encode simple algorithms into a high-level computer language (Python).</li> <li>3. Understand the basics of algorithmic parallelism and its importance for high-speed computing.</li> <li>4. Re-encode simple algorithms into parallelized, accelerated versions (also in Python).</li> <li>5. Master the parallel-processing capabilities of Python.</li> <li>6. Encode new and/or optimize existing computational problems, effectively reducing their simulation times from hours to seconds.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                 |     |
| <b>Education Method</b>                 | <ul style="list-style-type: none"> <li>- Lectures (including live coding)</li> <li>- Computer-lab assignments in the Cloud</li> <li>- Lab reports and final exam</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Computer Use</b>                     | Lab work will be performed on cutting-edge computers in the Cloud with NVidia graphics cards on board. Remote access to Cloud computers will be arranged annually as necessary to both offer enhanced lab flexibility and to familiarize students with the Cloud-computing paradigm. Regarding programming languages, Python knowledge is required for use in the labs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Books</b>                            | Lecture notes and supplementary materials will be provided during the lectures.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Prerequisites</b>                    | Python                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Assessment</b>                       | <p>Grading scheme: Pass/Fail.</p> <p>Lab reports are based on four lab assignments and contribute, in total, to 50% of the final mark. Lab assignments must be handed in on time, are graded independently and are compulsory to pass (as a whole).</p> <p>The final exam is a randomized multiple-choice test directly on Brightspace. It contributes to the remaining 50% of the final mark. The exam is graded independently and is compulsory to pass.</p> <p>The final mark is pass/fail and is calculated as the average of the lab-assignment mark and the final-exam mark. A student needs a final mark <math>\geq 5.8/10.0</math> to pass, as per standard TUDelft regulations.</p> <p>Exam resit: This is only possible for students who failed to acquire a passing mark in the final exam and/or (in rare cases) a passing mark in the lab-assignment mark. In such cases, an oral exam is, most commonly, arranged in the weeks after the final exam. This has to be arranged with the course instructor based on time availability.</p> |     |
| <b>Permitted Materials during Tests</b> | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| <b>Enrolment / Application</b>          | Yes, via Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |
| <b>Special Information</b>              | The teacher can be reached at c.strydis@erasmusmc.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
| <b>Elective</b>                         | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |
| <b>Tags</b>                             | Algorithmics<br>Computer Engineering<br>Modelling<br>Optimisation<br>Practicals<br>Programming<br>Programming concepts<br>Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
| <b>Studyload/Week</b>                   | 4h/week lecture/instruction, 8h/week of practical time available with TAs plus 4-12 h/week at home.<br><br>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder, requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take you.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Location</b>                         | Delft Campus and via remote access for part of the lab assignments.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |

| NB3017                                 | Quantum mechanics for Nanobiology 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2.5 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>          | T. Idema                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/8/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| <b>Education Period</b>                | 3<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Start Education</b>                 | 3<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Exam Period</b>                     | 3<br>3B<br>4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Required for</b>                    | NB3018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Expected prior knowledge</b>        | Physics 1A, 1B and 2, Analysis 1-3, Linear algebra, and Differential equations. The course will be fairly math-heavy, so a good understanding of the prerequisite math courses is essential.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Course Contents</b>                 | <p>This description applies to both part 1 (NB3017) and part 2 (NB3018) of Quantum mechanics for Nanobiology:</p> <p>In this course we introduce the concepts and applications of quantum mechanics in the context of nanobiology. We will cover basic quantum ideas (particle-wave duality, atomic orbitals and shells, tunneling) and put them on a solid mathematical foundation (part 1). Building on this foundation, we also explore atomic emission and absorption spectra, the fundamental principle behind lasers, the quantum mechanics behind chemistry, and properties of solids, especially conductors, semiconductors and insulators (part 2).</p> <p>The cornerstone of the course (and quantum mechanics) is the Schrödinger equation, together with its solutions (known as the wavefunctions) and the associated statistical interpretation. As the Schrödinger equation is a differential equation, and the wavefunctions are its eigenfunctions, this course combines what students have learned in their earlier physics and math courses. In explaining the periodic table and the mechanism behind photosynthesis, it also directly connects to the courses chemistry and physical biology of the cell.</p> |     |
| <b>Study Goals</b>                     | <p>At the completion of this course, the students will be able to:</p> <ul style="list-style-type: none"> <li>- Indicate the difference between quantum and classical dynamics.</li> <li>- Explain the concept of the quantum wave function and quantum operators, and perform mathematical operations with them.</li> <li>- Solve the one-dimensional Schrödinger equations for a number of common potentials, including square and harmonic wells and barriers.</li> <li>- Explain the idea behind the Heisenberg uncertainty principle, and apply it to arbitrary operators.</li> <li>- Solve the three-dimensional Schrödinger equations for the hydrogen atom.</li> <li>- Calculate emission spectra for atoms.</li> <li>- Calculate angular momenta in quantum mechanics.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Education Method</b>                | Lectures and tutorials. Video lectures covering the mathematical derivations, complemented by notes / sections of the book, will be provided each week. On-campus (or, if necessary, live online) sessions with the teachers will complement these lectures, focusing on concepts, with ample room for questions and working out example problems. Students work on problems themselves (with the help of TAs) during tutorials; some of these worked solutions are to be handed in each week and will be graded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
| <b>Books</b>                           | D. J. Griffiths and D. F. Schroeter, Introduction to quantum mechanics, third edition (2018), Cambridge University Press, ISBN 978-1-107-18963-8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
| <b>Assessment</b>                      | <p>Homework and final written exam; optional midterm. The final written exam will be combined with (and given at the same time as) that of NB3018. The final grade will consist for 25% of the homework and 75% of the exam grade; students can earn a bonus point for the exam with the optional midterm.</p> <p>Students taking NB3017 only (not NB3018) will take a modified final exam at the same time as students taking both courses. The grade will be calculated as homework+exam+bonus. Students should register for the NB3017 exam in Osiris.</p> <p>Students taking both NB3017 and NB3018 will take one final exam. The grade for NB3017 will be calculated as exam+bonus+NB3017 homework. The grade for NB3018 will be calculated as exam+bonus+NB3018 homework so the two grades may be different. Students should register for both the NB3017 and the NB3018 exam in Osiris.</p> <p>The retake will follow the same pattern as the regular exam. If there are very few participants, the retake may be an oral instead of a written exam.</p>                                                                                                                                                                    |     |
| <b>Remarks</b>                         | The course requires students to actively participate throughout, and work seriously during all sessions. In principle, students should attend all sessions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Studyload/Week</b>                  | <p>Preparation (video lecture / lecture notes) 2x2 hours, session with teacher 2x2 hours, tutorial 2x2 hours, finishing homework 1x2 hours = 14 hours per week.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Continuing Courses</b>              | NB3018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |

| NB3018                                 | Quantum mechanics for Nanobiology 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2.5 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>          | T. Idema                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/8/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| <b>Education Period</b>                | 3<br>3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Start Education</b>                 | 3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |
| <b>Exam Period</b>                     | 3<br>4<br>3B<br>4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Expected prior knowledge</b>        | Physics 1 and 2, Analysis 1-3, Linear algebra, and Differential equations. The course will be fairly math-heavy, so a good understanding of the prerequisite math courses is essential. This course immediately follows NB3017 (which is thus a key prerequisite). Obviously this part follows the course Quantum mechanics for Nanobiology 1 (NB3017).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Course Contents</b>                 | <p>This description applies to both part 1 (NB3017) and part 2 (NB3018) of Quantum mechanics for Nanobiology:</p> <p>In this course we introduce the concepts and applications of quantum mechanics in the context of nanobiology. We will cover basic quantum ideas (particle-wave duality, atomic orbitals and shells, tunneling) and put them on a solid mathematical foundation (part 1). Building on this foundation, we also explore atomic emission and absorption spectra, the fundamental principle behind lasers, the quantum mechanics behind chemistry, and properties of solids, especially conductors, semiconductors and insulators (part 2).</p> <p>The cornerstone of the course (and quantum mechanics) is the Schrödinger equation, together with its solutions (known as the wavefunctions) and the associated statistical interpretation. As the Schrödinger equation is a differential equation, and the wavefunctions are its eigenfunctions, this course combines what students have learned in their earlier physics and math courses. In explaining the periodic table and the mechanism behind photosynthesis, it also directly connects to the courses chemistry and physical biology of the cell.</p> |     |
| <b>Study Goals</b>                     | <p>At the completion of this course, the student will be able to:</p> <ul style="list-style-type: none"> <li>- Explain the concept of spin, identify fermionic and bosonic particles, and apply the Pauli exclusion principle.</li> <li>- Explain the shape of the periodic table.</li> <li>- Use perturbation theory and the variational principle to study the properties of quantum mechanical systems without known analytical solutions.</li> <li>- Explain the properties of conductors, insulators and semi-conductors with quantum mechanics.</li> <li>- Explain how molecular bonds form based on quantum mechanics, and use the LCAO (linear combination of atomic orbitals) to approximate the ground state energy of simple molecules.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| <b>Education Method</b>                | Lectures and tutorials. Video lectures covering the mathematical derivations, complemented by notes / sections of the book, will be provided each week. On-campus (or, if necessary, live online) sessions with the teachers will complement these lectures, focusing on concepts, with ample room for questions and working out example problems. Students work on problems themselves (with the help of TAs) during tutorials; some of these worked solutions are to be handed in each week and will be graded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
| <b>Books</b>                           | D. J. Griffiths and D. F. Schroeter, Introduction to quantum mechanics, third edition (2018), Cambridge University Press, ISBN 978-1-107-18963-8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
| <b>Prerequisites</b>                   | NB3017                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Assessment</b>                      | <p>Homework and final written exam; optional midterm (after NB3017). The final written exam will be combined with (and given at the same time as) that of NB3017. The final grade will consist for 25% of the homework and 75% of the exam grade; students can earn a bonus point for the exam with the optional midterm.</p> <p>Students taking NB3017 only (not NB3018) will take a modified final exam at the same time as students taking both courses. The grade will be calculated as homework+exam+bonus. Students should register for the NB3017 exam in Osiris.</p> <p>Students taking both NB3017 and NB3018 will take one final exam. The grade for NB3017 will be calculated as exam+bonus+NB3017 homework. The grade for NB3018 will be calculated as exam+bonus+NB3018 homework so the two grades may be different. Students should register for both the NB3017 and the NB3018 exam in Osiris.</p> <p>The retake will be organized the same way as the regular exam. If there are very few participants for the retake, it may be given as an oral instead of a written exam.</p>                                                                                                                                   |     |
| <b>Remarks</b>                         | The course requires students to actively participate throughout, and work seriously during all sessions. In principle, students should attend all sessions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Studyload/Week</b>                  | <p>Preparation (video lecture / lecture notes) 2x2 hours, session with teacher 2x2 hours, tutorial 2x2 hours, finishing homework 1x2 hours = 14 hours per week.</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |



| NB3019                                 | Molecular Motors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2.5 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>          | Prof.dr. N.H. Dekker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Education Period</b>                | 4<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Start Education</b>                 | 4<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Exam Period</b>                     | 4<br>4B<br>5AB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Expected prior knowledge</b>        | Completion of the first two years of study of the Nanobiology or the Physics programs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| <b>Course Contents</b>                 | This course provides an introduction to the molecular machines that govern many processes in cellular transport and motion. All biological examples will be studied with a view to the underlying physical principles                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Course Contents Continuation</b>    | The course encompasses integrated lecture and homework sessions and a final exam (3 hours).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Study Goals</b>                     | <p>At the completion of this course the student will be able to:</p> <p>Understand how the basic laws of mechanics, statistical mechanics, and fluid dynamics govern the behavior of nanoscale molecular machines.</p> <p>Understand the different types of possible couplings between chemical energy and mechanical motions</p> <p>Be able to describe the biological phenomena from a biophysical perspective, complete with the underlying mathematics.</p> <p>Analyze microscopic biological phenomena quantitatively.</p> <p>Apply the basic physical concepts learned here to current research articles.</p> |     |
| <b>Education Method</b>                | Lectures and discussion of research articles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Literature and Study Materials</b>  | Handouts will be provided in class.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |
| <b>Books</b>                           | Mechanics of Motor Proteins and the Cytoskeleton, by Jonathon Howard. This book is required material.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Assessment</b>                      | Homework (30%) and final exam (70%).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |

| NB3020                                 | Genomics Technology in Breast Cancer Research                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2.5 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>          | Prof. J.W.M. Martens                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/4*/4*/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |
| <b>Education Period</b>                | 3<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Start Education</b>                 | 3<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Exam Period</b>                     | 3<br>3A<br>3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Expected prior knowledge</b>        | Basic knowledge in oncology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Course Contents</b>                 | <p>The course Genomics Technology in Breast Cancer Research highlights state-of-the-art technological developments in the fields of genomics and how this can be applied to discover new biomarkers, exemplified for breast cancer. The course will be given in the format of classical lectures and will cover the introduction of (breast) cancer development, different types of genomic technological platforms as well as the bioinformatics and mathematics behind data analysis. The course will also include a journal club and students will present and discuss literature. During the course students will get a clearer understanding of how physics (technology development) and mathematics (data analysis) can be applied to relevant issues in cancer research. During the lectures, students understanding of the matter will be tested actively. During sessions students will read assigned literature and prepare a presentation and propose discussion points for the final journal club. This elective course is complementary to the elective course on Nanobiology in Cancer, but it is not compulsory to take the latter in advance.</p> |     |
| <b>Study Goals</b>                     | <p>A. Understand the principles of different technological platforms used in the fields of genomics.<br/> B. Acquire knowledge on which technological platform to use to answer clinically relevant biological questions in the context of (breast) cancer research</p> <p>At the completion of the course the students will be able to:</p> <ol style="list-style-type: none"> <li>1. Describe the basic principles of (breast) cancer genomics and understand the need for translational research</li> <li>2. Describe the different technological platforms available and understand the principles and its application</li> <li>3. Describe which technological platform will give an answer to which biological problem</li> <li>4. Find suitable background information related to the subjects of the lectures</li> <li>5. Critically read a scientific paper and present it to their peers</li> <li>6. Ask critical questions and discuss the value of reading assignments</li> </ol>                                                                                                                                                                     |     |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>- Lectures</li> <li>- Quizzes (optional)</li> <li>- Journal club</li> <li>- Presentations</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Assessment</b>                      | Graded course. Students will be evaluated based on presence at lectures (10%), presentation of assigned paper for the journal club (30%), and written examination and/or quizzes (60%).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Enrolment / Application</b>         | This course is offered twice (once in 3A and once in 3B) and has limited enrollment. Enrollment will be available after November 1 via the Nanobiology Bachelor Brightspace page and is done on a first come, first served basis with priority for Nanobiology students.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
| <b>Special Information</b>             | The teacher can be reached at: j.martens@erasmusmc.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Tags</b>                            | Group work<br>Lab Research<br>Research Methods<br>Software<br>Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
| <b>Location</b>                        | Erasmus MC, Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |

| NB3021                                 | Optics and its application in Nanobiology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2.5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>          | Dr. F. Bociort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Instructor</b>                      | Dr. E.C.M. Carroll                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| <b>Instructor</b>                      | Dr.ir. D. Brinks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Education Period</b>                | 3<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Start Education</b>                 | 3<br>3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Exam Period</b>                     | 3<br>3A<br>3B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Required for</b>                    | For students who intend to follow the Applied Physics Master Programme, this course can replace the course TN2345 in the Bridging Programme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Expected prior knowledge</b>        | The Optics part of this course is a continuation of the Optics part of the course NB2041 Optics and Microscopy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Course Contents</b>                 | Maxwell's equations and electromagnetic waves in vacuum and non-conducting materials, electric dipole radiation, dispersion, Huygens-Fresnel Principle, Fermat's Principle, reflection, refraction and transmission of electromagnetic waves at an interface between two media, Fresnel equations for reflectance and transmittance, total internal reflection and evanescent waves, fiber optics, optical tweezers and magnetic tweezers to study forces in biology, fluorescence lifetime imaging microscopy, Förster resonant energy transfer, optogenetics controlling cells with light, multiphoton microscopy, introduction to nonlinear optics, optics of short-pulsed lasers, coherent anti-Stokes Raman scattering microscopy |     |
| <b>Study Goals</b>                     | The student will have a good understanding of the basics of optics and several applications in nanobiology. The student will be able to answer questions and to apply the theory to solve simple problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Education Method</b>                | lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Literature and Study Materials</b>  | hand-outs of the lecture material<br>optional: E. Hecht, Optics, any edition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Assessment</b>                      | Written examination, both for the main exam and for the resit. Normally the exam is closed-book, off-campus exams are open-book.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |

| NB3022                                 | Epigenetics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2.5 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>          | Ir. H. Mira Bontenbal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Education Period</b>                | 4<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| <b>Start Education</b>                 | 4<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| <b>Exam Period</b>                     | 4<br>4A<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| <b>Expected prior knowledge</b>        | <p>Prior to the beginning of this course, students are expected to have knowledge of transcription/gene regulation, cell and developmental biology. Students should have completed course(s) Molecular Biology, Physical Biology of the Cell 1 and Evolutionary Developmental Biology.</p> <p>The course will assume knowledge of cellular biomolecules gained in the courses of Molecular Biology and Physical Biology of the Cell 1. Concepts of cell differentiation and development of Physical Biology of the Cell 1 and Evolutionary Developmental Biology are particularly important for this course. Knowledge from Molecular Biology about genes, transcription factors and chromatin are also required.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Course Contents</b>                 | <p>How is it that our bodies have hundreds of different cell types, each having very different patterns of gene expression and well-defined behaviours and functions, and yet all have the same genome? Why do the cells in your skin divide into two other skin cells but not into two neurons or two muscle cells? How is it possible that our diets will affect our children and even grandchildren? All these facts have something in common which is the heritability of information across cell generations, without modifications to the DNA sequence. They can be explained by epigenetics, which is the field that studies heritable changes in gene expression that do not involve changes in the DNA sequence.</p> <p>In this course, the students will deepen their understanding of the epigenetic mechanisms that mediate gene expression changes, development and disease. The course will be divided into three parts:</p> <ol style="list-style-type: none"> <li>1. The first part will be devoted to describing what the different epigenetic mechanisms are, such as histone modifications and DNA methylation, that regulate genes in general, but also at the level of two paradigms in epigenetics, imprinted genes and X chromosome inactivation. We will also highlight the role of nuclear organization in gene expression control.</li> <li>2. In the second part of this course, we will study how epigenetics affect development and disease, and we will look into epigenetic modulators that are used to combat disease. We will also discuss prions as a non-traditional epigenetic mechanism that can lead to disease.</li> <li>3. Lastly, the students will learn what the different techniques are that are currently available to study epigenetic modifications and their implications, from basic biology to the medical field.</li> </ol> <p>This elective course is very relevant to Nanobiologists who want to expand their knowledge of epigenetic control of gene expression, how epigenetics is critical to understanding cell differentiation and development, and some types of diseases and cancer progression, and what the current technologies are used to study epigenetics in basic biology, medical research and biotechnological companies.</p> |     |
| <b>Study Goals</b>                     | <p>At the completion of this course, the student will be able to:</p> <ul style="list-style-type: none"> <li>- Rephrase what epigenetics is.</li> <li>- Cite and explain the different types of epigenetic modifications that exist in mammals and what their mechanisms of action are in regulating gene expression.</li> <li>- Describe what imprinted genes and X chromosome inactivation are.</li> <li>- Explain how epigenetics can modulate disease and cancer progression.</li> <li>- Cite and explain the existing current technologies to study epigenetic mechanisms of gene expression control.</li> <li>- Rephrase essential concepts of epigenetics</li> <li>- Apply the obtained knowledge in order to critically describe, interpret and explain the results of scientific experiments in epigenetics, using provided text and figures from scientific manuscripts.</li> <li>- Read and critically assess scientific literature (soft skill).</li> <li>- Collaborate with fellow students to prepare a scientific presentation (soft skill).</li> <li>- Present a scientific argument, engage in discussion with fellow students and answer questions in public (soft skills)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>- Interactive lectures</li> <li>- Interactive practicals</li> <li>- Homework (graded)</li> <li>- Workshop: analysis of genome-wide data</li> <li>- Group work: students will work in group to prepare a presentation and present in front of their peers. There will be peer feedback, which will count for your presentation grade.</li> </ul> <p>Soft skills developed in this course: group work, presentation, peer review of presentation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Books</b>                           | <p>No book is required for this course. However, most of the figures in the lectures and practicals come from Epigenetics 2nd Edition (Allis, Caparros, Jenuwein, Reinberg, Lachner). All chapters of this book are open-access published reviews. A repository of the figures and their legends can be found at <a href="http://learnepigenetics.com/">http://learnepigenetics.com/</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Assessment</b>                      | <p>Assessment:</p> <ul style="list-style-type: none"> <li>20% Homework Assignments</li> <li>20% Oral group presentation</li> <li>60% Exam.</li> </ul> <p>(they can compensate each other)</p> <p>If the exam and re-exam are not passed, the grades for the homework and presentation are kept for next year.</p> <p>There is no resit for the Presentation. An exceptional alternative homework assignment will be given in case the student cannot do the regular homework assignment.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Special Information</b>             | <p>The teacher can be reached at: <a href="mailto:h.mirabontenbal@erasmusmc.nl">h.mirabontenbal@erasmusmc.nl</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |
| <b>Remarks</b>                         | <p>Hegias Mira Bontenbal can be reached at <a href="mailto:h.mirabontenbal@erasmusmc.nl">h.mirabontenbal@erasmusmc.nl</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Studyload/Week</b>                  | <p>2.5 ECTS<br/> 4h/week * 4 weeks = 16h in-class hours<br/> ca. 8h / week * 4 weeks = 32h study at home, homework, group work for presentation preparation<br/> 20h exam preparation</p> <p>Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| <b>Schedule</b>                        | <p>I will put the up-to-date schedule on Brightspace. So please follow that schedule and not the schedule on <a href="http://mytimetable.tudelft.nl">mytimetable.tudelft.nl</a>. Use the Brightspace one when the course starts.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |

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| <b>Location</b> | Erasmus MC, Rotterdam |
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|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>NB3023</b>                           | <b>Human Complex Genetics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>2.5</b> |
| <b>Responsible Instructor</b>           | Dr. J.G.J. van Rooij                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |
| <b>Instructor</b>                       | L. Broer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |
| <b>Education Period</b>                 | 4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |
| <b>Start Education</b>                  | 4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |
| <b>Exam Period</b>                      | 4A<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |
| <b>Expected prior knowledge</b>         | Basic knowledge of human genetics is recommended but not required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |
| <b>Course Contents</b>                  | <p>This course provides an introduction into the field of complex genetics. Also referred to as population genetics; generating large-scale genetic datasets and then answering relevant biological or clinical questions. The course is split in four contact days and exam:</p> <p>Day 1: Introduction to complex genetics (genetic variation, genome-wide association studies - GWAS, applications of complex genetics).<br/> Day 2: Interpretation of GWAS results (GWAS analysis methodology, biological interpretation of genetic variants).<br/> Day 3: Clinical application of GWAS results (disease risk prediction, polygenic risk scores, clinical decision-making).<br/> Day 4: Discussion of assignments; practical examples performed in groups or extracted from literature.</p> <p>Through these topics the course covers the basics of performing complex genetic studies, as well as some of the most common and currently utilized applications.</p> |            |
| <b>Study Goals</b>                      | <p>At completion of this course, the student will be able to:</p> <ul style="list-style-type: none"> <li>- Indicate the difference between clinical and complex/population genetics</li> <li>- Perform a complex genetic analysis (genome-wide association study)</li> <li>- Interpret complex genetic results in biological pathways</li> <li>- Calculate polygenic risk scores and apply them to predict disease risk</li> <li>- Indicate the (possible) role of genetics in medicine</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |
| <b>Education Method</b>                 | <p>The course is hosted in part by researchers from the Human Genotyping Facility (HuGe-F) at Erasmus MC. It focuses on the practical aspects of complex genetic studies and provides you hands-on experience and practical examples. Each course day contains a lecture (~1h) followed by a practical session on the topic of that day (first three course days). The practical sessions will be held using pre-prepared scripts in R. The students are invited to work on a practical example in small groups and discuss this in the 4th day. The course will develop professional skills; data analytics, interpretation of results, group work, presenting results.</p>                                                                                                                                                                                                                                                                                            |            |
| <b>Assessment</b>                       | Homework and practical assignment to be discussed on day 4. Individual exam on day 5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |
| <b>Permitted Materials during Tests</b> | Open book exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |
| <b>Remarks</b>                          | To contact the teacher, please use the EMC email address: j.vanrooij@erasmusmc.nl.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |
| <b>Studyload/Week</b>                   | This course stands for 2.5 ECTS, or approximately 14h/week. Approximately 4h each week is reserved for the lecture and practical sessions. The remaining time is intended for homework (week 1-3), preparing the group assignment (week 4) and preparing for the exam (week 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |
| <b>Location</b>                         | Rotterdam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |

| NB3024                                 | Advanced Math Topics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2.5 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Responsible Instructor</b>          | S.M.F. Abdoel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Instructor</b>                      | Dr. B. van den Dries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |
| <b>Instructor</b>                      | Prof.dr.ir. M.B. van Gijzen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Education Period</b>                | 4<br>4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| <b>Start Education</b>                 | 4A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |
| <b>Exam Period</b>                     | 4A<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| <b>Expected prior knowledge</b>        | Prerequisite courses are Differential Equations (NB2191) and Computational Science (NB2181).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Course Contents</b>                 | Many physical phenomena can be described by differential equations. In the context of Nanobiology you can think of processes like heat dispersion, diffusion or chemical reactions.<br>Often these equations have to be solved numerically. First the course will discuss how a system of ordinary differential equations (ODEs) can be solved using numerical time integration. Then we discuss how a stationary diffusion equation equation can be discretized with the Finite Difference Method. Finally we combine time integration and space discretisation to numerically solve the heat equation.<br><br>Lesson 1. Review of numerical differentiation. Some time integration methods: Forward Euler, Backward Euler, Trapezoidal Method, Modified Euler. Implicit methods and Explicit methods. Predictor-corrector methods.<br>Lesson 2. Stability of time integration methods. Order of accuracy. Systems of ODEs.<br>Lesson 3. Finite Difference Method. Application to 1D diffusion equation. Discretisation of boundary conditions. Properties of the discretisation matrix.<br>Lesson 4. Finite Difference Method (continued): Non-constant diffusion. Heat Equation, method of lines. |     |
| <b>Study Goals</b>                     | General: given an initial value problem or boundary value problem, the student can make a motivated choice of appropriate numerical method to find a numerical approximation of the solution, and analyse the quality of the approximation.<br><br>More specifically:<br><br>1. The student knows, can apply and can implement numerical time integration methods to solve ordinary differential equations (ODEs) and initial value problems, such as Forward Euler, Backward Euler, Trapezoidal Method and Modified Euler.<br>2. The student can classify methods as implicit method, explicit method and/or predictor-corrector method, and can analyse strengths and weaknesses.<br>3. The student can analyse stability and order of accuracy of time integration methods for (systems of) ODE(s).<br>4. The student can apply, implement and analyse the discretisation of the 1D diffusion equation with the Finite Difference Method.<br>5. The student can numerically solve the time-dependent 1D heat equation with the method of lines.                                                                                                                                                   |     |
| <b>Education Method</b>                | Four lecture classes and four instruction classes with programming assignments.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| <b>Literature and Study Materials</b>  | Numerical Methods for Ordinary Differential Equations<br>C. Vuik, F.J. Vermolen, M.B. van Gijzen and M.J. Vuik<br>Delft Academic Press, Delft, 2018<br>ISBN 978-90-6562-3737                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Assessment</b>                      | Project assignments plus oral exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
| <b>Tags</b>                            | Algorithmics<br>Lineair Algebra<br>Mathematics<br>Modelling<br>Numeric Methods<br>Practicals<br>Programming Software                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |
| <b>Location</b>                        | TU Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |

| NB3030                                 | Nanobiology Independent Research                                                                                                                                                                                                                                                                                                                                                                                                              | 0 |
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| <b>Responsible Instructor</b>          | Dr. J. Pothof                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Coordinator</b>              | J.C. Colgrove                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | variable                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | interdisciplinary independent research                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Study Goals</b>                     | At the end of the project, students will have learned to:<br>- develop feasible research questions<br>- applying innovative methods to the address these questions<br>- communicate about scientific work in an effective way<br>- write reports about their findings                                                                                                                                                                         |   |
| <b>Education Method</b>                | In this course you carry out an independent research project with the help of a scientific advisor from the program. If you would like to do such a project, you may join an existing project (an example would be the Corona Research super project spring of 2020) or propose one to the Programme director. Activities could include literature review, modeling, and lab work, and must include scientific discussion and report writing. |   |
| <b>Assessment</b>                      | Grading is Pass/Fail.<br>You will be assessed on your research log of work done and your final written report.                                                                                                                                                                                                                                                                                                                                |   |
| <b>Special Information</b>             | Starting an independent project must be discussed with the Nanobiology program and approved prior to beginning the project. The program may begin a project and open registration to any students at any time.                                                                                                                                                                                                                                |   |
| <b>Studyload/Week</b>                  | Credits are variable, students receive up to 1 ECTS per week based on at least 28 hours per week, with fractions thereof for less weekly effort. Credit requests with justification (log book and final report) must be sent to the responsible teacher who will assess the effort and register credits in Osiris.                                                                                                                            |   |

| TBM301A                                | Writing a Bachelors Thesis in English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2.5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Module Manager</b>                  | Drs. D.W. Laponder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |
| <b>Instructor</b>                      | S.F. Johnson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Instructor</b>                      | Drs. K.I. van der Linden                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>Instructor</b>                      | Drs. M.A. Swennen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
| <b>Co-responsible for assignments</b>  | S.F. Johnson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Contact Hours / Week</b><br>x/x/x/x | 2/0/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Education Period</b>                | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Start Education</b>                 | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>Course Contents</b>                 | This course is designed for students who are in the process of writing their Bachelors thesis. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly text analysis and writing assignments.                                                                                                  |     |
| <b>Study Goals</b>                     | The course is primarily designed for students working on their Bachelors thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure a paper, report or thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing. |     |
| <b>Education Method</b>                | Learning by doing and learning from corrections, lectures, assignments and tutor feedback.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| <b>Reader</b>                          | Course materials will be provided digitally.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| <b>Assessment</b>                      | Continuous assessment. Participants are to submit a number of text analysis and writing assignments in which they demonstrate their writing skills and their understanding of text composition and academic English. Students that do not submit their weekly assignments may be asked to leave the course.<br><br>An 85% attendance record is required. This means students may miss no more than one session (preferably none).<br><br>Students will receive their course credits when all conditions are met.                                   |     |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>Tags</b>                            | Diverse<br>English proficiency<br>Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |

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Dr.ir. W.M. Baarends

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L. Broer

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Dr.ir. S.J.J. Brouns

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Dr. W.A. van Cappellen

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| <b>Department</b> | Onderwijs en Studentenzaken    |

Dr. E.C.M. Carroll

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Dr. M.P. Chien

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J.C. Colgrove

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| <b>Department</b> | Onderwijs en Studentenzaken |
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Prof.dr. N.H. Dekker

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Dr.ir. M.W. Docter

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L. van der Elst

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Dr. H.J. Geertsema

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Prof.dr.ir. M.B. van Gijzen

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Dr. T.L.M. ten Hagen

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Prof.dr. A.B. Houtsmuller

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| <b>Department</b> | Onderwijs en Studentenzaken    |

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T. Idema

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Drs. E. Jacobs

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M.M. Jaegle

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S.F. Johnson

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Dr. J.A. van der Knaap

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Drs. D.W. Laponder

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Dr.ir. J.H.G. Lebbink

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| <b>Department</b> | Onderwijs en Studentenzaken    |

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Drs. K.I. van der Linden

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| <b>Unit</b>       | Technische Natuurwetenschappen |
| <b>Department</b> | Onderwijs en Studentenzaken    |

Ir. M.J.F. Negrello

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A. Pines

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| <b>Department</b> | Onderwijs en Studentenzaken |
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Dr. J. Pothof

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Prof.dr. F.H.J. Redig

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G. Roshchupkin

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| <b>Department</b> | Onderwijs en Studentenzaken |
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Dr. I.V. Smal

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Dr. I.M. Smit

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| <b>Department</b> | Bio-Electronics                |

Drs. M.A. Swennen

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Dr. B.C. Tilly

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| <b>Department</b> | Onderwijs en Studentenzaken    |
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Prof. W. Vermeulen

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|-------------------|-----------------------------|

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